Sampling Design and Survey Practice

Lecture Notes (Ch. 1–11)

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Chapter 1. Introduction and Elementary Concepts

1 Collection of Survey Data

1.1 Primary and secondary data

Definition 1.1

- **Primary data**: data collected by the investigator directly from the source.
- **Secondary data**: data that has already been collected by some agency or individual and is available in published or unpublished records.

Primary data fit the study's target; secondary data are cheaper but collected for another purpose. Primary data can be collected by

- direct personal interviews, or interviews by enumerators,
- questionnaires sent by mail,
- telephone interviews.

1.2 Framing a questionnaire

The person framing the questionnaire should know the field of inquiry in detail. Points to keep in mind:

- The surveyor introduces themselves and states the objective of the survey.
- Questions are clear, unambiguous and to the point; leading questions are avoided.
- Questions affecting the prestige or sentiments of respondents, and questions requiring calculations, are avoided.
- Precise instructions for filling in the questionnaire, including units of measurement, are given.
- The questionnaire is concise: keep the number of questions close to the number of quantities to be estimated.
- The layout is visually appealing.

1.3 Technical terms

Definition 1.2

- An **element** is a unit for which information is sought.
- The **population** (universe) is the aggregate of elements about which inference is to be made.
- **Sampling units** are non-overlapping collections of population elements.
- The **sampling frame** is a list of all sampling units in the population.
- A **sample** is a subset of the population selected from the frame to draw inferences about population characteristics.

Observing every unit is a **complete enumeration** or **census**. A census is often impossible, and a sample survey is faster, more detailed, cheaper and often more accurate (trained staff and follow-up reduce non-sampling errors).

1.4 Probability and non-probability sampling

The method used to select the sample is the **sampling procedure**.

- **Probability sampling**: units are selected by a probability mechanism. Only these procedures are considered in this course.
- **Non-probability sampling**: units are selected without a probability mechanism (convenience, quota, judgement); such procedures are subject to human bias.

1.5 With and without replacement

Definition 1.3

- **With replacement (WR)**: each drawn unit is returned before the next draw; with order, N^n possible samples.
- **Without replacement (WOR)**: the selected unit is not returned before the next draw. There are $\binom{N}{n}$ possible (unordered) samples.

Example 1.4: Birth weights

Four children A, B, C, D have birth weights $5.5, 8.0, 6.5, 7.0$ pounds. The WR samples of size 2 (ordered) are the $4^2 = 16$ pairs

$$(A,A),(A,B),(A,C),(A,D),(B,A),\ldots,(D,D),$$

whereas WOR gives the $\binom{4}{2} = 6$ samples $\{A,B\},\{A,C\},\{A,D\},\{B,C\},\{B,D\},\{C,D\}$.

In R: `sample(n)` is a random permutation of $1,\ldots,n$; `sample(x)` permutes x; `sample(x, n)` draws n items WOR; `sample(x, n, replace=T)` draws WR; `sample(x, replace=T)` is a bootstrap sample.

1.6 Planning and execution of a sample survey

Objectives → population to be studied → sampling unit → sampling frame → sample selection (sample size and procedure) → method of collecting information → pilot survey → handling of nonresponse → organization of fieldwork → analysis of data and report.

2 Elementary Concepts

2.1 Expectation, variance, covariance

If x takes values $x_1,\ldots,x_k$ with probabilities $p_1,\ldots,p_k$,

$$\mathrm{E}(x) = \sum_{i=1}^{k} p_i x_i, \qquad \mathrm{Var}(x) = \mathrm{E}[x - \mathrm{E}(x)]^2 = \mathrm{E}(x^2) - [\mathrm{E}(x)]^2,$$

$$\mathrm{Cov}(x,y) = \mathrm{E}[(x-\mathrm{E}x)(y-\mathrm{E}y)] = \mathrm{E}(xy) - \mathrm{E}(x)\mathrm{E}(y),$$
$$\mathrm{Var}\Big(\sum_{i=1}^{k} a_i x_i\Big) = \sum_{i=1}^{k} a_i^2 \mathrm{Var}(x_i) + 2\sum_{i<j} a_i a_j \mathrm{Cov}(x_i, x_j).$$

2.2 Parameters and statistics

Definition 1.5

A **parameter** is a real function of all population values, a **statistic** a real function of the sample values; a statistic estimating a parameter is an **estimator**.

For population values $Y_1, \ldots, Y_N$ and sample values $y_1, \ldots, y_n$:

$$\mu = \frac{1}{N}\sum_{i=1}^{N} Y_i, \quad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(Y_i-\mu)^2, \quad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(Y_i-\mu)^2 = \frac{N}{N-1}\sigma^2,$$
$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2.$$

The finite-population variance S^2 (divisor $N-1$) is the natural target of s^2 under SRS without replacement (Ch. 2), and $(N-1)S^2 = N\sigma^2$.

2.3 Bias, variance and MSE

Definition 1.6

- $\hat{\theta}$ is **unbiased** for θ if $\mathrm{E}(\hat{\theta}) = \theta$. Otherwise its **bias** is $B(\hat{\theta}) = \mathrm{E}(\hat{\theta}) - \theta$ and its **relative bias** is $RB(\hat{\theta}) = B(\hat{\theta})/\theta$.
- Sampling variance $\mathrm{Var}(\hat{\theta}) = \mathrm{E}[\hat{\theta} - \mathrm{E}\hat{\theta}]^2$; standard error $SE(\hat{\theta}) = \sqrt{\mathrm{Var}(\hat{\theta})}$.
- Mean squared error $\mathrm{MSE}(\hat{\theta}) = \mathrm{E}(\hat{\theta} - \theta)^2$.
- The **relative efficiency** of $\hat{\theta}_2$ with respect to $\hat{\theta}_1$ is $\mathrm{RE}(\hat{\theta}_1, \hat{\theta}_2) = \mathrm{MSE}(\hat{\theta}_1)/\mathrm{MSE}(\hat{\theta}_2)$; if it exceeds 1, $\hat{\theta}_2$ is more efficient.

Theorem 1.7: Bias–variance decomposition

$\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + [B(\hat{\theta})]^2$.

Proof.

$$\begin{aligned}\mathrm{E}(\hat{\theta}-\theta)^2 &= \mathrm{E}\big[(\hat{\theta} - \mathrm{E}\hat{\theta}) + (\mathrm{E}\hat{\theta} - \theta)\big]^2 \\ &= \mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta})^2 + (\mathrm{E}\hat{\theta} - \theta)^2 + 2(\mathrm{E}\hat{\theta} - \theta)\,\mathrm{E}(\hat{\theta} - \mathrm{E}\hat{\theta}),\end{aligned}$$

and the last expectation is zero. ∎

2.4 Confidence intervals and the bound on the error

For a desired probability $1-\alpha$ we look for a bound B with

$$P(|\hat{\theta} - \theta| \le B) = 1-\alpha \iff P(\hat{\theta} - B \le \theta \le \hat{\theta} + B) = 1-\alpha,$$

so $(\hat{\theta} - B, \hat{\theta} + B)$ is a confidence interval with confidence coefficient $1-\alpha$.

- **Normal case**: for (approximately) normal unbiased $\hat{\theta}$, $B = z_{\alpha/2}\sigma_{\hat{\theta}}$; at 95%, $z_{0.025} \approx 2$ gives the "2-SD bound" $B = 2\sigma_{\hat{\theta}}$ used throughout.
- **Any distribution**: by Chebyshev's inequality $P(|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}) \ge 1 - \frac{1}{4} = 0.75$.

2.5 Sample size determination

The required precision $P(|\hat{\theta} - \theta| \le B) = 1 - \alpha$ is set by the investigator, with **tolerable error** (margin of error) B; for proportions in surveys of people, typically $B = 0.03$, $\alpha = 0.05$.

Example 1.8: Opinion polls

With the normal approximation and $p = 1/2$ (the worst case),

$$B = 1.96\sqrt{\frac{p(1-p)}{n}} \le 1.96\sqrt{\frac{1}{4n}}.$$

Setting $B = 0.03$ gives $n = (1.96)^2/(4{\cdot}0.03^2) = 1067.1$, so about 1,068 respondents; this is why national polls typically report a margin of error of $\pm 3\%$ with about a thousand respondents.

3 Selected exercises

Motivation. The 2-SD guarantee, an exact sampling distribution by enumeration, and a biased estimator beating an unbiased one in MSE.

Exercise 1.1 (Chebyshev guarantee for the bound on the error)

Let $\hat{\theta}$ be an estimator of θ with finite variance, and let $k > 0$.
(a) If $\hat{\theta}$ is unbiased with standard error $\sigma_{\hat{\theta}} > 0$, show that $P(|\hat{\theta} - \theta| \le k\sigma_{\hat{\theta}}) \ge 1 - 1/k^2$.
(b) Without assuming unbiasedness, show that $P(|\hat{\theta} - \theta| \le k\sqrt{\mathrm{MSE}(\hat{\theta})}) \ge 1 - 1/k^2$ whenever $\mathrm{MSE}(\hat{\theta}) > 0$, and explain why the 2-SD bound of a biased estimator should be built from $\sqrt{\mathrm{MSE}}$ rather than from the standard error.

Solution. [Supplement] Both parts follow from one inequality.

Step 1. (Markov's inequality.) For a random variable $U \ge 0$ and $a > 0$, $a\,\mathbf{1}\{U \ge a\} \le U$ pointwise; taking expectations gives $P(U \ge a) \le \mathrm{E}(U)/a$.

Step 2. Apply Step 1 to $U = (\hat{\theta} - \theta)^2$ and $a = k^2\mathrm{E}(\hat{\theta} - \theta)^2 = k^2\mathrm{MSE}(\hat{\theta}) > 0$:

$$P(|\hat{\theta} - \theta| > k\sqrt{\mathrm{MSE}(\hat{\theta})}) \le P((\hat{\theta} - \theta)^2 \ge k^2\mathrm{MSE}(\hat{\theta})) \le \frac{\mathrm{MSE}(\hat{\theta})}{k^2\mathrm{MSE}(\hat{\theta})} = \frac{1}{k^2}.$$

Taking complements proves (b).

(a) If $\mathrm{E}\hat{\theta} = \theta$, $\mathrm{MSE}(\hat{\theta}) = \sigma^2_{\hat{\theta}}$, so (a) is a special case of (b); $k = 2$ gives at least $3/4$.

For biased $\hat{\theta}$, $\mathrm{MSE} = \sigma^2_{\hat{\theta}} + B^2$, and $|\hat{\theta} - \theta| \le 2\sigma_{\hat{\theta}}$ may have probability below $1/2$ (normal $\hat{\theta}$, $|B| > 2\sigma_{\hat{\theta}}$); use $\sqrt{\mathrm{MSE}}$. ■

Exercise 1.2 (Birth weights: exact sampling distribution of $\bar{y}$)

For the birth weights $5.5, 8.0, 6.5, 7.0$ (example on WR and WOR sampling), let $\bar{y}$ be the mean of a sample of size $n = 2$.
(a) Under WOR, list the six sample means and compute $\mathrm{E}(\bar{y})$ and $\mathrm{Var}(\bar{y})$ by enumeration.
(b) Under WR (ordered pairs, each with probability $1/16$), compute $\mathrm{Var}(\bar{y})$ and the ratio of the two variances. Compare with σ^2/n and with $\frac{N-n}{N-1}\frac{\sigma^2}{n}$.

Solution. The population mean and variance are $\mu = 27/4 = 6.75$ and

$$\sigma^2 = \tfrac{1}{4}\left[(-1.25)^2 + 1.25^2 + (-0.25)^2 + 0.25^2\right] = \tfrac{3.25}{4} = 0.8125.$$

(a) The six samples, each with probability $1/6$, have means

$\{A,B\}$	$\{A,C\}$	$\{A,D\}$	$\{B,C\}$	$\{B,D\}$	$\{C,D\}$
6.75	6.00	6.25	7.25	7.50	6.75

Their sum is 40.5, so $\mathrm{E}(\bar{y}) = 40.5/6 = 6.75 = \mu$. The squared deviations from 6.75 are 0, 0.5625, 0.25, 0.25, 0.5625, 0, so

$$\mathrm{Var}(\bar{y}) = \frac{1.625}{6} = 0.2708.$$

(b) Under WR the draws are independent and uniform, so $\mathrm{Var}(\bar{y}) = \sigma^2/2 = 0.40625$ (as enumeration of the 16 ordered pairs confirms). The ratio is

$$\frac{0.40625}{0.2708} = 1.5 = \frac{N-1}{N-n} = \frac{3}{2}.$$

So WR gives σ^2/n and WOR $\frac{N-n}{N-1}\frac{\sigma^2}{n} = \frac{2}{3}\cdot 0.40625$ (Ch. 2): WOR is 50% more efficient here, observing half the population. ■

Exercise 1.3 (Shrinking the sample mean)

Let $y_1, \ldots, y_n$ be iid with mean $\mu \neq 0$ and variance $\sigma^2 > 0$, and consider $\hat{\theta}_c = c\,\bar{y}$ for a constant c.
(a) Find $\mathrm{MSE}(\hat{\theta}_c)$ as a function of c.
(b) Find the minimizing c^* and show that $\mathrm{MSE}(\hat{\theta}_{c^*}) < \mathrm{MSE}(\bar{y})$.
(c) Compute $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*})$ and discuss why $\hat{\theta}_{c^*}$ is not used in practice.

Solution. [Supplement]

(a) $\mathrm{E}(\hat{\theta}_c) = c\mu$ and $\mathrm{Var}(\hat{\theta}_c) = c^2\sigma^2/n$, so by the bias–variance decomposition

$$\mathrm{MSE}(c) = c^2\frac{\sigma^2}{n} + (c-1)^2\mu^2.$$

(b) $\mathrm{MSE}(c)$ is a quadratic in c with leading coefficient $\sigma^2/n + \mu^2 > 0$, and $\mathrm{MSE}'(c) = 2c\sigma^2/n + 2(c-1)\mu^2 = 0$ gives

$$c^* = \frac{\mu^2}{\mu^2 + \sigma^2/n} \in (0,1).$$

Substituting, with $v = \sigma^2/n$,

$$\mathrm{MSE}(c^*) = \frac{\mu^4 v + v^2\mu^2}{(\mu^2+v)^2} = \frac{\mu^2 v}{\mu^2 + v} < v = \mathrm{MSE}(\bar{y}),$$

the inequality being strict because $v > 0$.

(c) $\mathrm{RE}(\bar{y}, \hat{\theta}_{c^*}) = (\mu^2 + v)/\mu^2 = 1 + \sigma^2/(n\mu^2) > 1$: shrinkage wins, especially when σ/μ is large relative to $\sqrt{n}$. But c^* involves the unknown σ^2/μ^2, and a plug-in c loses the guarantee of (b). Unbiasedness is a convention; MSE is the criterion. ■

Chapter 2. Simple Random Sampling

1 Introduction

Definition 2.1

In **simple random sampling** (SRS) every sample of size n is equally likely; without replacement each of the $\binom{N}{n}$ samples has probability $1/\binom{N}{n}$.

A simple random sample can be drawn by the lottery method or with random number tables / computer programs (`sample(N, n)` in R).

Setting. The population values are $u_1, \dots, u_N$ with

$$\mu = \frac{1}{N}\sum_{i=1}^{N} u_i, \qquad \sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(u_i-\mu)^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}(u_i-\mu)^2,$$

and $y_1, \dots, y_n$ is an SRS drawn without replacement (WOR). The **sampling fraction** is $f = n/N$ and $1-f$ is the **finite population correction** (fpc).

2 Estimation of the Population Mean

Motivation. The only randomness is which units are selected; the SRS mean is unbiased with finite population correction $(N-n)/N$.

Theorem 2.2: Sample mean under SRS (WOR)

Let $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ and $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i-\bar{y})^2$. Then

(i) $\mathrm{E}(\bar{y}) = \mu$;

(ii) $\mathrm{Var}(\bar{y}) = \dfrac{N-n}{N-1}\dfrac{\sigma^2}{n} = \dfrac{N-n}{N}\dfrac{S^2}{n} = \dfrac{S^2}{n}(1-f)$;

(iii) $\mathrm{E}(s^2) = S^2$, hence $\widehat{\mathrm{Var}}(\bar{y}) = \dfrac{s^2}{n}(1-f)$ is unbiased for $\mathrm{Var}(\bar{y})$.

Derivation: proof of the theorem

Marginal distribution of one draw. By symmetry each y_i equals u_k with probability $1/N$, so

$$\mathrm{E}(y_i) = \sum_{k=1}^{N} u_k \frac{1}{N} = \mu, \qquad \mathrm{Var}(y_i) = \sum_{k=1}^{N}(u_k-\mu)^2\frac{1}{N} = \sigma^2 \quad \text{for all } i.$$

(i) follows: $\mathrm{E}(\bar{y}) = \frac{1}{n}\sum \mathrm{E}(y_i) = \mu$.

Covariance of two draws. For $i \neq j$ the pair (y_i, y_j) is (u_k, u_l), $k \neq l$, with probability $\frac{1}{N(N-1)}$. Hence

$$\mathrm{Cov}(y_i, y_j) = \mathrm{E}(y_i y_j) - \mu^2 = \frac{1}{N(N-1)}\sum_{k\neq l} u_k u_l - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2$$

$$= \frac{1}{N(N-1)}\Big[\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] - \frac{1}{N^2}\Big(\sum_k u_k\Big)^2$$
$$= \frac{1}{N(N-1)}\Big[\frac{1}{N}\Big(\sum_k u_k\Big)^2 - \sum_k u_k^2\Big] = -\frac{1}{N(N-1)}\Big(\sum_k u_k^2 - N\mu^2\Big)$$
$$= -\frac{1}{N(N-1)}\sum_k (u_k-\mu)^2 = -\frac{\sigma^2}{N-1}.$$

Variance of the mean.

$$\mathrm{Var}(\bar y) = \frac{1}{n^2}\Big[\sum_i \mathrm{Var}(y_i) + 2\sum_{i<j}\mathrm{Cov}(y_i,y_j)\Big] = \frac{1}{n^2}\Big[n\sigma^2 - n(n-1)\frac{\sigma^2}{N-1}\Big]$$
$$= \frac{\sigma^2}{n}\Big(1 - \frac{n-1}{N-1}\Big) = \frac{\sigma^2}{n}\frac{N-n}{N-1} = \frac{S^2}{n}\frac{N-n}{N},$$

using $N\sigma^2 = (N-1)S^2$.

Unbiasedness of s^2. Using $\sum(y_i-\bar y)^2 = \sum(y_i-\mu)^2 - n(\bar y-\mu)^2$,

$$\mathrm{E}(s^2) = \frac{1}{n-1}[n\sigma^2 - n\mathrm{Var}(\bar y)] = \frac{\sigma^2}{n-1}\Big(n - \frac{N-n}{N-1}\Big) = \frac{\sigma^2}{n-1}\cdot\frac{N(n-1)}{N-1} = \frac{N}{N-1}\sigma^2 = S^2.$$

Therefore $\mathrm{E}[\widehat{\mathrm{Var}}(\bar y)] = \frac{N-n}{Nn}\mathrm{E}(s^2) = \mathrm{Var}(\bar y)$.

Remark 2.3: SRS with replacement

If the draws are made with replacement, the y_i are iid with mean μ and variance σ^2. Then $\mathrm{Cov}(y_i,y_j) = 0$ (e.g. for 0/1 values $\mathrm{E}(y_iy_j) = P(y_i=1)P(y_j=1) = \frac{N_1}{N}\frac{N_1}{N}$), so

$$\mathrm{Var}(\bar y) = \frac{\sigma^2}{n}, \qquad \mathrm{E}(s^2) = \frac{1}{n-1}\Big(n\sigma^2 - n\frac{\sigma^2}{n}\Big) = \sigma^2, \qquad \widehat{\mathrm{Var}}(\bar y) = \frac{s^2}{n}.$$

Without replacement the variance is smaller by the factor $\frac{N-n}{N-1}$; the fpc matters only when f is not small.

Confidence interval and bound. A $100(1-\alpha)\%$ confidence interval for μ is

$$\bar y \pm z_{\alpha/2}\sqrt{1-f}\,\frac{s}{\sqrt{n}},$$

and with $\alpha = 0.05$ we use the 2-SD bound $B = 2\sqrt{\widehat{\mathrm{Var}}(\bar y)} = 2\sqrt{\frac{s^2}{n}(1-f)}$.

Example 2.4: Tractors per village

Estimate the average number of tractors per village; $N = 69$, $n = 10$, $\bar y = 16.7$, $s^2 = 81.344$.

$$\widehat{\mathrm{Var}}(\bar y) = \frac{s^2}{n}\cdot\frac{N-n}{N} = \frac{81.344}{10}\cdot\frac{59}{69} = 6.956, \qquad 2\sqrt{6.956} = 5.27,$$

so the interval is $16.7 \pm 5.27 = [11.43,\ 21.97]$.

3 Estimation of the Population Total

Motivation. The total is N times the mean, with N^2 times its variance.

The population total is $\tau = N\mu$, and with N known the natural estimator is $\hat{\tau} = N\bar{y}$.

Theorem 2.5

$\mathrm{E}(\hat{\tau}) = \tau$ and

$$\mathrm{Var}(\hat{\tau}) = N^2 \frac{N-n}{N-1}\frac{\sigma^2}{n} = N^2 \frac{N-n}{N}\frac{S^2}{n}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2 \frac{N-n}{N}\frac{s^2}{n} = N^2 \frac{s^2}{n}(1-f).$$

A $100(1-\alpha)\%$ interval is $\hat{\tau} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$, typically $\hat{\tau} \pm 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau})}$.

Example 2.6: Total number of tractors

$\hat{\tau} = 69 \times 16.7 = 1152.3$, $\widehat{\mathrm{Var}}(\hat{\tau}) = 69^2 \times 6.956 = 33{,}117$, and $2\sqrt{33117} = 364.0$. The interval is $1152.3 \pm 364.0 = [788, 1516]$.

4 Sample Size for Estimating μ and τ

To estimate μ with a bound B, set $2\sqrt{\mathrm{Var}(\bar{y})} = B$:

$$B = 2\sqrt{\frac{N-n}{N-1}\frac{\sigma^2}{n}} \implies n = \frac{N\sigma^2}{(N-1)D + \sigma^2} = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4}.$$

Derivation

Write $D = B^2/4$. Then $D\,n(N-1) = (N-n)\sigma^2$, i.e. $n[(N-1)D + \sigma^2] = N\sigma^2$. The second form follows from $\sigma^2 = \frac{N-1}{N}S^2$.

For the total, $B = 2\sqrt{N^2\frac{N-n}{N}\frac{S^2}{n}}$ gives

$$n = \frac{NS^2}{ND + S^2}, \qquad D = \frac{B^2}{4N^2}.$$

In practice S^2 (or σ^2) is unknown and replaced by a prior estimate or a pilot-sample s^2.

Example 2.7: Tractors, continued

With $s^2 = 81.344$ and $B = 3$ ($D = 2.25$),

$$n = \frac{69 \times 81.344}{69 \times 2.25 + 81.344} = \frac{5612.7}{236.6} = 23.72 \;\Rightarrow\; n = 24,$$

so 14 more villages are needed beyond the 10 already sampled.

5 Estimation of a Population Proportion

With $y_i = 1$ if unit i has the characteristic, 0 otherwise, $p = \mu$ and

$$\hat{p} = \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i, \qquad \hat{q} = 1 - \hat{p}.$$

For 0/1 values $\sigma^2 = pq$ and $S^2 = \frac{N}{N-1}pq$.

Theorem 2.8: Proportion under SRS

$$\mathrm{E}(\hat{p}) = p, \qquad \mathrm{Var}(\hat{p}) = \frac{pq}{n}\frac{N-n}{N-1}, \qquad \widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N} = \frac{\hat{p}\hat{q}}{n-1}(1-f),$$

and $\mathrm{E}[\widehat{\mathrm{Var}}(\hat{p})] = \mathrm{Var}(\hat{p})$.

Derivation: why the divisor is $n-1$

$$\mathrm{E}(\hat{p}\hat{q}) = \mathrm{E}(\hat{p}) - \mathrm{E}(\hat{p}^2) = p - [\mathrm{Var}(\hat{p}) + p^2] = pq - \frac{pq}{n}\frac{N-n}{N-1}$$
$$= pq\,\frac{n(N-1)-(N-n)}{n(N-1)} = \frac{N(n-1)}{(N-1)n}\,pq.$$

Hence

$$\mathrm{E}\Big[\frac{\hat{p}\hat{q}}{n-1}\frac{N-n}{N}\Big] = \frac{N-n}{N(n-1)}\cdot\frac{N(n-1)}{(N-1)n}pq = \frac{pq}{n}\frac{N-n}{N-1} = \mathrm{Var}(\hat{p}).$$

By contrast the plug-in $\frac{\hat{p}\hat{q}}{n}(1-f)$ is biased downward: $\mathrm{E}\big[\frac{\hat{p}\hat{q}}{n}(1-f)\big] \neq \mathrm{Var}(\hat{p})$. This is the same phenomenon as s^2 versus the divisor-n variance, since $s^2 = \frac{n}{n-1}\hat{p}\hat{q}$ for 0/1 data.

Example 2.9: Student vote

An SRS of $n = 120$ from $N = 2400$ students; 72 support candidate A. Then $\hat{p} = 0.6$ and

$$\widehat{\mathrm{Var}}(\hat{p}) = \Big(1 - \frac{120}{2400}\Big)\frac{0.6\times 0.4}{119} = 0.001916, \qquad 2\sqrt{0.001916} = 0.0875,$$

so the interval is $0.6 \pm 0.0875 = (0.512,\ 0.688)$.

6 Sample Size for Estimating p

Replace σ^2 by pq in the formula for μ:

$$n = \frac{Npq}{(N-1)D + pq}, \qquad D = \frac{B^2}{4} \quad (\text{or } D = B^2/1.96^2 \text{ with the exact normal quantile}).$$

With no prior information take $p = 1/2$, which maximizes pq and gives a conservative n.

Example 2.10

For $N = 2400$ and $B = 0.05$: $D = 0.05^2/4 = 0.000625$ and

$$n = \frac{2400\times 0.25}{2399\times 0.000625 + 0.25} = \frac{600}{1.7494} = 342.98 \ \Rightarrow\ n = 343.$$

7 Towards Stratified Sampling

Consider estimating the total first-hospitalization cost of kidney-stone patients. Two regions with very different incidence rates are surveyed separately:

	Carolinas	Rockies
n	363	258
$\bar{y}$	1350	1150

Pooling all 621 observations ignores the separate regional samples and population sizes; weight each regional mean by its population size (the stratified estimator, Chapter 3).

8 Selected exercises

Motivation. The SRS variance from inclusion indicators, the covariance of two sample means (used in Ch. 5), and a proportion computation.

Exercise 2.1 (Inclusion indicators)

Draw an SRS of size n without replacement from units $1, \dots, N$ with values $u_1, \dots, u_N$, and let $Z_k = 1$ if unit k is in the sample and $Z_k = 0$ otherwise.
(a) Show that $P(Z_k = 1) = n/N$ and, for $k \neq l$, $P(Z_k = Z_l = 1) = \frac{n(n-1)}{N(N-1)}$.
(b) Writing $\bar{y} = \frac{1}{n}\sum_{k=1}^{N} Z_k u_k$, rederive $\mathrm{E}(\bar{y}) = \mu$ and $\mathrm{Var}(\bar{y}) = \frac{S^2}{n}(1-f)$.

Solution. [Supplement]

(a) The samples containing k choose the other $n-1$ of $N-1$ units: $P(Z_k = 1) = \binom{N-1}{n-1}/\binom{N}{n} = n/N$; similarly $P(Z_k Z_l = 1) = \binom{N-2}{n-2}/\binom{N}{n} = \frac{n(n-1)}{N(N-1)}$.

(b)

Step 1. (Moments of the indicators.) By (a), $\mathrm{E}Z_k = f$, $\mathrm{Var}Z_k = f(1-f)$ (as $Z_k^2 = Z_k$), and for $k \neq l$

$$\mathrm{Cov}(Z_k, Z_l) = \frac{n(n-1)}{N(N-1)} - \frac{n^2}{N^2} = \frac{n}{N}\Big(\frac{n-1}{N-1} - \frac{n}{N}\Big) = -\frac{f(1-f)}{N-1}.$$

Step 2. (Mean.) Only the Z_k are random, so $\mathrm{E}\bar{y} = \frac{1}{n}\sum_k f\, u_k = \frac{1}{N}\sum_k u_k = \mu$.

Step 3. (Variance.) Since $\sum_k Z_k = n$ is constant, $\bar{y} - \mu = \frac{1}{n}\sum_k Z_k(u_k - \mu)$; write $d_k = u_k - \mu$, so $\sum_k d_k = 0$ and $\sum_k d_k^2 = (N-1)S^2$. Then

$$n^2\mathrm{Var}(\bar{y}) = \sum_k d_k^2 \mathrm{Var}Z_k + \sum_{k\neq l} d_k d_l \mathrm{Cov}(Z_k, Z_l) = f(1-f)\Big[\sum_k d_k^2 - \frac{1}{N-1}\sum_{k\neq l} d_k d_l\Big].$$

Step 4. Because $\sum_{k\neq l} d_k d_l = \big(\sum_k d_k\big)^2 - \sum_k d_k^2 = -\sum_k d_k^2$, the bracket equals $\sum_k d_k^2\big(1 + \frac{1}{N-1}\big) = \frac{N}{N-1}(N-1)S^2 = NS^2$. Hence

$$\mathrm{Var}(\bar{y}) = \frac{f(1-f)NS^2}{n^2} = \frac{S^2}{n}(1-f).$$

The same computation with arbitrary inclusion probabilities π_k leads to the Horvitz–Thompson estimator $\sum_k Z_k u_k/\pi_k$ of τ. ■

Exercise 2.2 (Covariance of two sample means)

Each unit carries two values (x_k, y_k), $k = 1, \dots, N$, with population means μ_x, μ_y and

$$S_{xy} = \frac{1}{N-1} \sum_{k=1}^{N} (x_k - \mu_x)(y_k - \mu_y).$$

For an SRS of n pairs without replacement, show that
(a) $\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n} S_{xy}$;
(b) $s_{xy} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$ satisfies $\mathrm{E}(s_{xy}) = S_{xy}$.

Solution. [Supplement] Apply the theorem on the sample mean under SRS to the auxiliary variable $z_k = x_k + y_k$, whose sample mean is $\bar{z} = \bar{x} + \bar{y}$.

(a)

Step 1. Expanding the square, $S_z^2 = \frac{1}{N-1} \sum_k [(x_k - \mu_x) + (y_k - \mu_y)]^2 = S_x^2 + S_y^2 + 2S_{xy}$.

Step 2. By the theorem, $\mathrm{Var}(\bar{z}) = \frac{1-f}{n} S_z^2$, $\mathrm{Var}(\bar{x}) = \frac{1-f}{n} S_x^2$ and $\mathrm{Var}(\bar{y}) = \frac{1-f}{n} S_y^2$.

Step 3. Since $\mathrm{Var}(\bar{z}) = \mathrm{Var}(\bar{x}) + \mathrm{Var}(\bar{y}) + 2\mathrm{Cov}(\bar{x}, \bar{y})$,

$$2\mathrm{Cov}(\bar{x}, \bar{y}) = \frac{1-f}{n} (S_z^2 - S_x^2 - S_y^2) = \frac{1-f}{n} \, 2S_{xy}.$$

(b) The sample analogue of Step 1 is $s_z^2 = s_x^2 + s_y^2 + 2s_{xy}$. Taking expectations and using $\mathrm{E}(s_z^2) = S_z^2$, $\mathrm{E}(s_x^2) = S_x^2$, $\mathrm{E}(s_y^2) = S_y^2$ from the theorem,

$$2\mathrm{E}(s_{xy}) = S_z^2 - S_x^2 - S_y^2 = 2S_{xy}.$$

Polarization turns SRS variance statements into covariance statements (used for the ratio estimator bias, Ch. 5). ■

Exercise 2.3 (Hunters of game birds)

An SRS of $n = 1000$ of the $N = 99{,}000$ licensed hunters of a province is taken, and 430 of them hunt game birds.
(a) Estimate the proportion p of hunters seeking game birds and place a bound on the error of estimation.
(b) How large a sample is needed to estimate p with a bound $B = 0.02$, using the estimate of (a) as prior value?

Solution.

(a) $\hat{p} = 430/1000 = 0.43$ and, by the proportion theorem,

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.43 \times 0.57}{999} \cdot \frac{98{,}000}{99{,}000} = 0.000243, \qquad 2\sqrt{0.000243} = 0.031,$$

so p is estimated by 0.43 ± 0.031.

(b) With $D = B^2/4 = 0.0001$ and $pq = 0.43 \times 0.57 = 0.2451$,

$$n = \frac{Npq}{(N-1)D + pq} = \frac{99{,}000 \times 0.2451}{98{,}999 \times 0.0001 + 0.2451} = \frac{24{,}265}{10.145} = 2391.8,$$

so $n = 2392$ hunters are needed. Since N is large, this is close to the infinite-population value $pq/D = 2451$; the fpc saves about 60 interviews. ■

Chapter 3. Stratified Random Sampling

1 Introduction

Definition 3.1

A **stratified random sample** splits the population into non-overlapping **strata** and draws independent SRSs within them.

Each stratum is a separate population. Stratification pays when the variation *between* strata is large: SRS models $y_i = \mu + \varepsilon_i$, stratified sampling $y_{ij} = \mu_j + \varepsilon_{ij}$. Design questions:

- how many strata to construct,
- how to allocate the total sample size to the strata,
- how to choose the strata.

2 Notation

Strata $j = 1, \dots, L$, units i, observations y_{ij}; SRS without replacement within strata.

N_j, n_j	population and sample size of stratum j; $N = \sum N_j$, $n = \sum n_j$
$w_j = N_j/N$	stratum weight
$f_j = n_j/N_j$	sampling fraction in stratum j
$\mu_j = \frac{1}{N_j}\sum_{i=1}^{N_j} Y_{ij}$	true stratum mean
$\bar{y}_j = \frac{1}{n_j}\sum_{i=1}^{n_j} y_{ij}$	sample stratum mean
$\sigma_j^2 = \frac{1}{N_j}\sum_i (Y_{ij} - \mu_j)^2$	true stratum variance
$S_j^2 = \frac{N_j}{N_j - 1}\sigma_j^2$	stratum variance with divisor $N_j - 1$
$s_j^2 = \frac{1}{n_j - 1}\sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2$	sample stratum variance

3 Estimation of the Population Mean

Motivation. Sampling each group separately removes between-group variation; under proportional allocation it never loses to SRS.

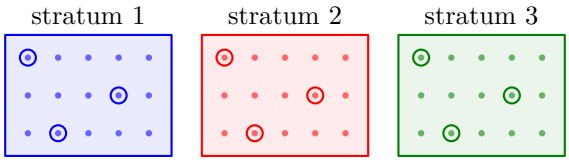


Figure. *Independent SRSs (circled) within homogeneous strata; only within-stratum variation enters* $\mathrm{Var}(\bar{y}_{st})$, $\bar{y}_{st} = \sum_h W_h \bar{y}_h$.

Theorem 3.2: Stratified estimator of μ

$$\bar{y}_{\mathrm{st}} = \frac{1}{N}\sum_{j=1}^{L} N_j \bar{y}_j = \sum_{j=1}^{L} w_j \bar{y}_j$$

is unbiased for μ, with

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}}) = \sum_{j=1}^{L} w_j^2 (1-f_j)\frac{s_j^2}{n_j}, \qquad \mathrm{E}[\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}})] = \mathrm{Var}(\bar{y}_{\mathrm{st}}).$$

A $100(1-\alpha)\%$ interval is $\bar{y}_{\mathrm{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}})}$.

Proof. Each $\bar{y}_j$ is an SRS mean, so by Ch. 2, $\mathrm{E}(\bar{y}_j) = \mu_j$, $\mathrm{Var}(\bar{y}_j) = (1-f_j)S_j^2/n_j$ and $\mathrm{E}(s_j^2) = S_j^2$. Thus

$$\mathrm{E}(\bar{y}_{\mathrm{st}}) = \sum_j w_j \mu_j = \frac{1}{N}(N_1\mu_1 + \cdots + N_L\mu_L) = \frac{1}{N}(\tau_1 + \cdots + \tau_L) = \frac{\tau}{N} = \mu.$$

Since the strata are sampled independently the covariances vanish and $\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \mathrm{Var}(\bar{y}_j)$. Unbiasedness of $\widehat{\mathrm{Var}}$ follows from $\mathrm{E}(s_j^2) = S_j^2$. ■

4 Estimation of the Population Total

Motivation. Stratum totals add, and so do their variances (independent samples).

Theorem 3.3

$\hat{\tau}_{\mathrm{st}} = N\bar{y}_{\mathrm{st}} = \sum_{j=1}^{L} N_j \bar{y}_j$ is unbiased for τ, and

$$\mathrm{Var}(\hat{\tau}_{\mathrm{st}}) = N^2 \mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{S_j^2}{n_j}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{st}}) = \sum_{j=1}^{L} N_j^2 (1-f_j)\frac{s_j^2}{n_j}.$$

Example 3.4: TV-viewing time

	N_j	w_j	n_j	$\bar{y}_j$	s_j
Town A	155	0.5	20	33.90	5.95
Town B	62	0.2	8	25.12	15.25
Rural	93	0.3	12	19.00	9.36
Total	310		40		

Mean.

$$\bar{y}_{\mathrm{st}} = \frac{155(33.90) + 62(25.12) + 93(19.00)}{310} = \frac{8578.94}{310} = 27.67.$$

$$\begin{aligned}\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{st}}) &= (0.5)^2\frac{5.95^2}{20}\left(1-\frac{20}{155}\right) + (0.2)^2\frac{15.25^2}{8}\left(1-\frac{8}{62}\right) + (0.3)^2\frac{9.36^2}{12}\left(1-\frac{12}{93}\right)\\ &= 0.385 + 1.013 + 0.572 = 1.97,\end{aligned}$$

so the estimate with a 2-SD bound is $27.67 \pm 2\sqrt{1.97} = 27.67 \pm 2.81$.
Total. $\hat{\tau}_{\mathrm{st}} = 310(27.67) = 8579$, $\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{st}}) = 310^2(1.97) \approx 189{,}400$, and the bound is $2\sqrt{189{,}400} \approx 870$.

5 Sample Size for Estimating μ and τ

Setting $2\sqrt{\mathrm{Var}(\bar{y}_{\mathrm{st}})} = B$ alone cannot be solved for n; we must also fix how n is split among the strata. Write the allocation as $n_j = na_j$ with $\sum a_j = 1$.

Theorem 3.5: Sample size for a given allocation

$$n = \frac{\sum_{j=1}^{L} N_j^2 S_j^2 / a_j}{N^2 D + \sum_{j=1}^{L} N_j S_j^2}, \qquad D = \begin{cases} B^2/4 & \text{for } \mu, \\ B^2/(4N^2) & \text{for } \tau, \end{cases}$$

with S_j^2 replaced by prior estimates s_j^2 in practice.

Derivation

For μ let $D = (B/2)^2 = \mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}(1 - \frac{n_j}{N_j})$. Multiply by N^2 and substitute $n_j = na_j$:

$$N^2 D = \sum_j N_j^2 \frac{S_j^2}{n a_j} - \sum_j N_j S_j^2 \implies n\Big(N^2 D + \sum_j N_j S_j^2\Big) = \sum_j \frac{N_j^2 S_j^2}{a_j}.$$

For τ, $\mathrm{Var}(\hat{\tau}_{\mathrm{st}}) = N^2 \mathrm{Var}(\bar{y}_{\mathrm{st}}) = B^2/4$ gives the same formula with $D = B^2/(4N^2)$. (Using 1.96 instead of 2 replaces $D = (B/2)^2$ by $(B/1.96)^2$.)

Example 3.6: Equal allocation

For the TV population ($N_j = 155, 62, 93$) with prior variances $s_1^2 = 25$, $s_2^2 = 225$, $s_3^2 = 100$, equal allocation $a_j = 1/3$ and $B = 2$ for μ ($D = 1$):

$$n = \frac{3(155^2 \cdot 25 + 62^2 \cdot 225 + 93^2 \cdot 100)}{310^2(1) + (155 \cdot 25 + 62 \cdot 225 + 93 \cdot 100)} = \frac{6{,}991{,}275}{123{,}225} = 56.7,$$

so $n = 57$ and $n_1 = n_2 = n_3 = 19$.

6 Allocation of the Sample

Let the survey cost be

$$TC = c_0 + \sum_{j=1}^{L} c_j n_j,$$

where c_0 is the overhead cost and c_j the cost per observation in stratum j.

6.1 Equal allocation

$n_j = n/L$. Then $TC = c_0 + \frac{n}{L}\sum c_j$, so $n = \dfrac{(TC - c_0)L}{\sum_j c_j}$.

6.2 Proportional allocation

$n_j = nw_j = nN_j/N$, i.e. $a_j = w_j$ and all $f_j = f$ are equal. Then $TC = c_0 + n\sum c_j w_j$, so $n = \dfrac{TC - c_0}{\sum_j c_j w_j}$; if $c_j \equiv c$, $n = (TC - c_0)/c$. The estimator becomes self-weighting: $\bar{y}_{\text{st}}$ equals the plain mean of all n observations.

6.3 Optimum allocation

Theorem 3.7: Optimum allocation

For a fixed total cost (equivalently, for a fixed variance) $\text{Var}(\bar{y}_{\text{st}})$ is minimized when

$$n_j \propto \frac{w_j S_j}{\sqrt{c_j}}, \qquad \text{i.e.} \qquad n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}},$$

and the total sample size under a budget TC is

$$n = \frac{(TC - c_0)\sum_k w_k S_k/\sqrt{c_k}}{\sum_k w_k S_k \sqrt{c_k}}.$$

Derivation: Cauchy–Schwarz argument

Split the variance into the part depending on the allocation and a constant:

$$\text{Var}(\bar{y}_{\text{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \underbrace{\sum_j \frac{w_j^2 S_j^2}{n_j}}_{V'} - \underbrace{\sum_j \frac{w_j S_j^2}{N}}_{\text{const}}.$$

Let $C' = TC - c_0 = \sum_j c_j n_j$. Minimizing V' for fixed C', or C' for fixed V', is the same as minimizing the product

$$V'C' = \Big(\sum_j \frac{w_j^2 S_j^2}{n_j}\Big)\Big(\sum_j c_j n_j\Big) = \Big(\sum_j a_j^2\Big)\Big(\sum_j b_j^2\Big), \quad a_j = \frac{w_j S_j}{\sqrt{n_j}},\ b_j = \sqrt{c_j n_j}.$$

By the Cauchy–Schwarz inequality

$$V'C' \geq \Big(\sum_j a_j b_j\Big)^2 = \Big(\sum_j w_j S_j \sqrt{c_j}\Big)^2,$$

whose right-hand side does not depend on the n_j. Equality holds iff b_j/a_j is constant:

$$\frac{b_j}{a_j} = \frac{n_j\sqrt{c_j}}{w_j S_j} = \text{const} \iff n_j \propto \frac{w_j S_j}{\sqrt{c_j}}.$$

Substituting $n_j = n\,\frac{w_j S_j/\sqrt{c_j}}{\sum_k w_k S_k/\sqrt{c_k}}$ into $TC = c_0 + \sum c_j n_j$ gives $TC = c_0 + n\alpha$ with $\alpha = \frac{\sum_k w_k S_k \sqrt{c_k}}{\sum_k w_k S_k/\sqrt{c_k}}$, so $n = (TC - c_0)/\alpha$.

We therefore take a larger sample in a stratum that is (1) larger, (2) more variable, or (3) cheaper to sample.

6.4 Neyman allocation

Theorem 3.8: Neyman allocation

If the costs are equal, $c_j \equiv c$, the optimum allocation becomes

$$n_j = n\,\frac{w_j S_j}{\sum_k w_k S_k} = n\,\frac{N_j S_j}{\sum_k N_k S_k}.$$

Example 3.9: Neyman allocation for the TV data

With $n = 40$ and $N_j s_j = 922.3,\ 945.5,\ 870.5$ (sum 2738.2), the Neyman allocation is

$$n_1 = 13.5, \quad n_2 = 13.8, \quad n_3 = 12.7 \ \Rightarrow\ (13, 14, 13),$$

versus the proportional $(20, 8, 12)$ used: the small, very variable Town B gets many more. With s_j for S_j, $\mathrm{Var}(\bar{y}_{\mathrm{st}})$ drops from 1.97 to about 1.66.

7 Estimation of a Population Proportion

Motivation. Proportions are means of 0–1 variables, with $p_h(1-p_h)$ as stratum variances.

With $y_{ij} \in \{0, 1\}$ the stratum mean $\bar{y}_j$ is the stratum proportion $\hat{p}_j$, and

$$\hat{p}_{\mathrm{st}} = \frac{1}{N}(N_1\hat{p}_1 + \cdots + N_L\hat{p}_L) = \sum_j w_j \hat{p}_j, \qquad \mathrm{E}(\hat{p}_{\mathrm{st}}) = \sum_j w_j p_j = p.$$

Theorem 3.10

$$\mathrm{Var}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{p_j q_j}{n_j}\Big(\frac{N_j - n_j}{N_j - 1}\Big), \qquad \widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}}) = \sum_j w_j^2 \frac{\hat{p}_j \hat{q}_j}{n_j - 1}(1 - f_j),$$

and a $100(1-\alpha)\%$ interval for p is $\hat{p}_{\mathrm{st}} \pm z_{\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{p}_{\mathrm{st}})}$.

These follow from Ch. 2 applied stratum by stratum, the divisor $n_j - 1$ making each term unbiased.

8 Selected exercises

Motivation. Exact gains of Neyman over proportional allocation and of proportional over SRS, and a design with unequal costs.

Exercise 3.1 (Proportional versus Neyman allocation)

With equal costs and fixed n, let $\bar{y}_{\mathrm{prop}}$ and $\bar{y}_{\mathrm{opt}}$ be the stratified means under $n_j = nw_j$ and Neyman $n_j = nw_jS_j/\bar{S}$, $\bar{S} = \sum_j w_j S_j$ ($n_j \le N_j$). Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\sum_{j=1}^{L} w_j (S_j - \bar{S})^2,$$

so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) > \mathrm{Var}(\bar{y}_{\mathrm{opt}})$ unless all S_j are equal.

Solution.

Step 1. (A common constant.) For any allocation,

$$\mathrm{Var}(\bar{y}_{\mathrm{st}}) = \sum_j w_j^2 \frac{S_j^2}{n_j}\Big(1 - \frac{n_j}{N_j}\Big) = \sum_j \frac{w_j^2 S_j^2}{n_j} - \frac{1}{N}\sum_j w_j S_j^2,$$

since $w_j^2/N_j = w_j/N$. The last sum does not depend on the allocation.

Step 2. (Proportional.) With $n_j = n w_j$, $\sum_j w_j^2 S_j^2/n_j = \frac{1}{n}\sum_j w_j S_j^2$.

Step 3. (Neyman.) With $n_j = n w_j S_j/\bar{S}$, $\sum_j w_j^2 S_j^2/n_j = \frac{\bar{S}}{n}\sum_j w_j S_j = \frac{\bar{S}^2}{n}$.

Step 4. (Difference.) Subtracting, the constants of Step 1 cancel and

$$\mathrm{Var}(\bar{y}_{\mathrm{prop}}) - \mathrm{Var}(\bar{y}_{\mathrm{opt}}) = \frac{1}{n}\Big(\sum_j w_j S_j^2 - \bar{S}^2\Big) = \frac{1}{n}\sum_j w_j (S_j - \bar{S})^2,$$

the last step being the identity $\sum_j w_j (S_j - \bar{S})^2 = \sum_j w_j S_j^2 - 2\bar{S}\sum_j w_j S_j + \bar{S}^2 \sum_j w_j = \sum_j w_j S_j^2 - \bar{S}^2$ (recall $\sum_j w_j = 1$).

Step 5. The right side is a positively weighted variance of the S_j, zero iff all $S_j = \bar{S}$ (then the allocations coincide): Neyman gains when strata differ in variability. ■

Exercise 3.2 (Proportional allocation versus SRS)

Let $\bar{y}_{\mathrm{ran}}$ be the mean of an SRS of size n from the whole population, with variance $\frac{1-f}{n}S^2$. Show that

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) - \mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \frac{1-f}{n(N-1)}\Big[\sum_{j=1}^{L} N_j(\mu_j - \mu)^2 - \sum_{j=1}^{L}(1 - w_j)S_j^2\Big],$$

and conclude that proportional allocation is better than SRS unless the stratum means are nearly equal.

Solution. [Supplement]

Step 1. (Analysis of variance of the population.) Adding and subtracting μ_j inside stratum j, the cross term vanishes because $\sum_i (Y_{ij} - \mu_j) = 0$:

$$(N-1)S^2 = \sum_j \sum_i (Y_{ij} - \mu)^2 = \sum_j (N_j - 1)S_j^2 + \sum_j N_j(\mu_j - \mu)^2.$$

Step 2. Under proportional allocation $f_j = f$, so $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \sum_j w_j^2 (1-f)\frac{S_j^2}{n w_j} = \frac{1-f}{n}\sum_j w_j S_j^2$. Hence

$$\mathrm{Var}(\bar{y}_{\mathrm{ran}}) - \mathrm{Var}(\bar{y}_{\mathrm{prop}}) = \frac{1-f}{n}\Big(S^2 - \sum_j w_j S_j^2\Big).$$

Step 3. By Step 1,

$$S^2 - \sum_j w_j S_j^2 = \frac{\sum_j N_j(\mu_j - \mu)^2}{N-1} + \sum_j S_j^2\Big(\frac{N_j - 1}{N-1} - \frac{N_j}{N}\Big),$$

and $\frac{N_j - 1}{N-1} - \frac{N_j}{N} = \frac{N(N_j-1) - N_j(N-1)}{N(N-1)} = -\frac{N - N_j}{N(N-1)} = -\frac{1-w_j}{N-1}$. Substituting gives the claim.

Step 4. The first sum is the between-strata sum of squares, growing with N_j; the second sums L within-stratum variances with coefficients below 1. Unless the μ_j are nearly equal, $\mathrm{Var}(\bar{y}_{\mathrm{prop}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$; for large N_j the difference is about $\frac{1-f}{n}\sum_j w_j(\mu_j - \mu)^2 \geq 0$. ■

Exercise 3.3 (Allocation with unequal costs)

A county estimates the proportion of families who will use an expanded day-care facility. Strata: city users, country users, city nonusers, country nonusers, $N_j = 97, 43, 145, 68$; expected proportions 0.9 (users) and 0.5 (nonusers); costs 4 (user) and 8 (nonuser). Find n and the allocation for $B = 0.05$.

Solution. For 0/1 data $S_j^2 = \frac{N_j}{N_j - 1} p_j q_j \approx p_j q_j$, so we use $p_j q_j = 0.09, 0.09, 0.25, 0.25$ in place of S_j^2.

Step 1. (Allocation.) By the optimum-allocation theorem, $n_j \propto w_j S_j / \sqrt{c_j} \propto N_j \sqrt{p_j q_j / c_j}$:

$$N_j \sqrt{p_j q_j / c_j} = 14.55,\ 6.45,\ 25.63,\ 12.02, \qquad \text{sum } 58.65,$$

so $a_j = 0.248,\ 0.110,\ 0.437,\ 0.205$.

Step 2. (Sample size.) With $N = 353$, $D = B^2/4 = 0.000625$ and the sample-size theorem,

$$n = \frac{\sum_j N_j^2 p_j q_j / a_j}{N^2 D + \sum_j N_j p_j q_j} = \frac{22{,}595}{77.88 + 65.85} = \frac{22{,}595}{143.73} = 157.2,$$

where $\sum_j N_j^2 p_j q_j / a_j = 846.8/0.248 + 166.4/0.110 + 5256.3/0.437 + 1156.0/0.205$ and $\sum_j N_j p_j q_j = 8.73 + 3.87 + 36.25 + 17.00$.

Step 3. $n = 158$ gives $n_j = 39.2,\ 17.4,\ 69.0,\ 32.4$, i.e. $(39, 17, 69, 33)$. Nonusers get more despite higher cost: $pq = 0.25$ vs 0.09, only partly offset by $1/\sqrt{c_j}$. ■

Chapter 4. Systematic Sampling

1 Introduction

Picking names at equal intervals from a random start gives a systematic sample.

Definition 4.1

A **1-in-k systematic sample with a random start** takes a random one of the first k frame elements and every kth thereafter; n is the sample size, k the number of possible samples.

Advantages over SRS:

- the sample is easy to draw and the field work is less prone to mistakes;
- it can provide more information per unit cost, because the sample is spread evenly over the frame.

2 Selection Methods

2.1 Method A: random start among the first k

Example 4.2: $N = 12$, $k = 3$

Picking y_2 among y_1, y_2, y_3 gives y_2, y_5, y_8, y_{11}: the population is cut into $n = N/k = 4$ rows, one unit per row, and the $k = 3$ samples are the columns

#1	#2	#3
y_1	y_2	y_3
y_4	y_5	y_6
y_7	y_8	y_9
y_{10}	y_{11}	y_{12}

In general the sample is $j,\ j+k,\ j+2k,\ \ldots,\ j+(n-1)k$ with j uniform on $\{1,\ldots,k\}$, and each systematic sample has probability $1/k$.

If $N \neq nk$ the sizes differ: $N = 12$, $k = 5$ give (y_1, y_6, y_{11}), (y_2, y_7, y_{12}), (y_3, y_8), (y_4, y_9), (y_5, y_{10}), each with probability $1/5$.

2.2 Method B: random start among all N

Select j at random from the whole population; with remainder r of j by k, start at y_r ($r = 0$: y_k) and take every kth unit.

Example 4.3: $N = 11$, $k = 3$

If $j = 8$, then $8 = 2 \cdot 3 + 2$, so $r = 2$, the start is y_2 and the sample is (y_2, y_5, y_8, y_{11}). The possible samples are

$$\#1 = (y_1, y_4, y_7, y_{10}), \quad \#2 = (y_2, y_5, y_8, y_{11}), \quad \#3 = (y_3, y_6, y_9).$$

Sample #2 is obtained whenever $j \in \{2, 5, 8, 11\}$, so its probability is $4/11$; similarly #1 has probability $4/11$ and #3 has $3/11$.

Remark 4.4: A versus B

Same samples, different probabilities: $1/k$ (Method A) vs n_i/N (Method B, needing N); equal when $N = nk$.

3 Estimation of the Population Mean

Let y_{ij} be the jth element of the ith systematic sample ($i = 1, \dots, k$, $j = 1, \dots, n_i$), $\bar{y}_i = \frac{1}{n_i}\sum_j y_{ij}$, and $\bar{y}_{\text{sy}}$ the mean of the sample actually selected.

3.1 Unbiasedness

Case 1 (Method A, $N = nk$).

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{k}(\bar{y}_1 + \cdots + \bar{y}_k) = \frac{1}{k}\frac{1}{n}(y_1 + \cdots + y_N) = \mu,$$

so $\bar{y}_{\text{sy}}$ is unbiased.

Example 4.5

Population $6, 3, 4, 9, 2, 5, 1, 7, 8$ with $k = 3$: the samples are $(6, 9, 1)$, $(3, 2, 7)$, $(4, 5, 8)$ with means $16/3$, $12/3$, $17/3$, whose average is $5 = \mu$.

Case 2 (Method A, $N \neq nk$). For $y_1, \dots, y_7$ and $k = 3$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{1}{3}\left[\frac{y_1 + y_4 + y_7}{3} + \frac{y_2 + y_5}{2} + \frac{y_3 + y_6}{2}\right] \neq \frac{y_1 + \cdots + y_7}{7} = \mu$$

in general: $\bar{y}_{\text{sy}}$ is *biased*.

Case 3 (Method B, $N \neq nk$). With probabilities $3/7, 2/7, 2/7$,

$$\mathrm{E}(\bar{y}_{\text{sy}}) = \frac{3}{7}\cdot\frac{y_1 + y_4 + y_7}{3} + \frac{2}{7}\cdot\frac{y_2 + y_5}{2} + \frac{2}{7}\cdot\frac{y_3 + y_6}{2} = \frac{y_1 + \cdots + y_7}{7} = \mu.$$

In general $\mathrm{E}(\bar{y}_{\text{sy}}) = \sum_i \frac{n_i}{N}\bar{y}_i = \frac{1}{N}\sum_i\sum_j y_{ij} = \mu$: Method B is always unbiased.

3.2 Variance of $\bar{y}_{\text{sy}}$ ($N = nk$)

By definition,

$$\mathrm{Var}(\bar{y}_{\text{sy}}) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_i - \mu)^2,$$

where $S^2 = \frac{1}{N-1}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \mu)^2$.

Theorem 4.6: Variance of the systematic mean

$$\mathrm{Var}(\bar{y}_{\text{sy}}) = \frac{N-1}{N}S^2 - \frac{1}{N}\sum_{i=1}^{k}\sum_{j=1}^{n}(y_{ij} - \bar{y}_i)^2.$$

Proof. Decompose the total sum of squares:

$$\sum_i\sum_j(y_{ij} - \mu)^2 = \sum_i\sum_j(y_{ij} - \bar{y}_i)^2 + n\sum_i(\bar{y}_i - \mu)^2 + 2\sum_i(\bar{y}_i - \mu)\sum_j(y_{ij} - \bar{y}_i),$$

and the cross term vanishes because $\sum_j (y_{ij} - \bar{y}_i) = 0$. The left side is $(N-1)S^2$ and $n\sum_i (\bar{y}_i - \mu)^2 = nk\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = N\mathrm{Var}(\bar{y}_{\mathrm{sy}})$. Divide by N. ■

The first term is the population variance, the second the pooled variation *within* systematic samples: heterogeneous samples make $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small.

Remark 4.7: Periodicity

For 1 2 3 4 5 | 1 2 3 4 5 | 1 2 3 4 5, $k = 5$, every sample is constant, e.g. $(2, 2, 2)$: no within-sample variation, maximal $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$.

3.3 Intraclass correlation

Homogeneity within a systematic sample is measured by the **intraclass correlation coefficient**

$$\rho = \frac{\mathrm{E}[(y_{ij} - \mu)(y_{ij'} - \mu)]}{\mathrm{E}[(y_{ij} - \mu)^2]}, \qquad j \neq j',$$

where the expectations are averages over the $kn(n-1)/2$ pairs in the same sample and over the N units, respectively.

Theorem 4.8

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{S^2}{n} \cdot \frac{N-1}{N}[1 + (n-1)\rho], \qquad \rho = \frac{2}{(n-1)(N-1)S^2} \sum_{i=1}^{k} \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu).$$

Derivation

$$\begin{aligned}\mathrm{Var}(\bar{y}_{\mathrm{sy}}) &= \frac{1}{k}\sum_i \Big(\frac{1}{n}\sum_j y_{ij} - \mu\Big)^2 = \frac{1}{n^2 k}\sum_i \Big(\sum_j (y_{ij} - \mu)\Big)^2 \\ &= \frac{1}{n^2 k}\Big[\sum_i \sum_j (y_{ij} - \mu)^2 + 2\sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu)\Big].\end{aligned}$$

The denominator of ρ is $\frac{1}{N}\sum_i \sum_j (y_{ij} - \mu)^2 = \frac{N-1}{N}S^2$, and the numerator is $\frac{2}{kn(n-1)}\sum_i \sum_{j<j'}(\cdot)(\cdot)$, so

$$2\sum_i \sum_{j<j'} (y_{ij} - \mu)(y_{ij'} - \mu) = kn(n-1)\frac{N-1}{N}S^2\rho = (n-1)(N-1)S^2\rho,$$

which is the stated formula for ρ. Substituting, with $n^2 k = nN$,

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{nN}[(N-1)S^2 + (n-1)(N-1)S^2\rho] = \frac{S^2}{n}\frac{N-1}{N}[1 + (n-1)\rho].$$

Compare with SRS, $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{S^2}{n}(1 - \frac{n}{N})$:

- ρ large and positive (homogeneous systematic samples) $\Rightarrow$ $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ large;
- ρ small positive or negative (heterogeneous samples) $\Rightarrow$ $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ small;
- $\rho \approx 0 \Rightarrow \mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$. Indeed $\rho = -\frac{1}{N-1}$ reproduces the SRS variance exactly.

3.4 ANOVA form

Arrange the population as an $n \times k$ table whose columns are the systematic samples, with column means $\bar{y}_1, \ldots, \bar{y}_k$ and grand mean μ. Let

$$SS_T = \sum_i \sum_j (y_{ij} - \mu)^2, \quad SS_W = \sum_i \sum_j (y_{ij} - \bar{y}_i)^2, \quad SS_B = n \sum_i (\bar{y}_i - \mu)^2,$$

so $SS_T = SS_W + SS_B$ and $S^2 = SS_T/(N-1)$. Then

$$\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{k} \sum_i (\bar{y}_i - \mu)^2 = \frac{SS_B}{N},$$

and equating with the ρ form, $\frac{SS_T}{nN}[1 + (n-1)\rho] = \frac{SS_B}{N}$, gives

$$1 + (n-1)\rho = \frac{n\, SS_B}{SS_T} \quad \Longrightarrow \quad \rho = 1 - \frac{n}{n-1}\frac{SS_W}{SS_T} \approx 1 - \frac{SS_W}{SS_T} = \frac{SS_B}{SS_T}.$$

A large between-sample sum of squares means a large ρ and a large variance.

Example 4.9: Two orderings of $1, \ldots, 9$

Here $\mu = 5$, $S^2 = 60/8 = 7.5$, $n = k = 3$, and the SRS variance is $\mathrm{Var}(\bar{y}_{\mathrm{ran}}) = \frac{7.5}{3} \cdot \frac{6}{9} = 1.667$.

- Order 1 2 3 | 4 5 6 | 7 8 9: the samples are $(1,4,7), (2,5,8), (3,6,9)$ with means $4, 5, 6$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(1+0+1) = \frac{2}{3}$ and $\rho = -21/60$. Heterogeneous samples: systematic beats SRS.
- Order 1 4 7 | 2 5 8 | 3 6 9: the samples are $(1,2,3), (4,5,6), (7,8,9)$ with means $2, 5, 8$. Then $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) = \frac{1}{3}(9+0+9) = 6$ and $\rho = 51/60$. Homogeneous samples: systematic is much worse than SRS.

Check: $\frac{7.5}{3} \cdot \frac{8}{9}(1 + 2\rho) = \frac{20}{9}(1 + 2\rho)$ equals $\frac{2}{3}$ for $\rho = -0.35$ and 6 for $\rho = 0.85$.

3.5 Types of population

We want systematic samples with heterogeneous units (small ρ). Which populations produce them?

1. **Random order**, e.g. weights of $N = 3000$ freshmen from an alphabetical list: $\rho \approx 0$, $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) \approx \mathrm{Var}(\bar{y}_{\mathrm{ran}})$; treat as SRS.
2. **Ordered population**, e.g. farms by area for corn yield: the sample covers the range, $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) < \mathrm{Var}(\bar{y}_{\mathrm{ran}})$, and is more representative than an SRS.
3. **Periodic variation**, e.g. supermarket sales by weekday: if k matches the period, ρ is large and $\mathrm{Var}(\bar{y}_{\mathrm{sy}}) > \mathrm{Var}(\bar{y}_{\mathrm{ran}})$.

3.6 Estimating the variance

One systematic sample carries no between-sample information, so $\mathrm{Var}(\bar{y}_{\mathrm{sy}})$ has no unbiased estimator; assuming random order ($\rho \approx 0$), use the SRS formula:

$$\bar{y}_{\mathrm{sy}} = \frac{1}{n} \sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{sy}}) = \frac{s^2}{n} \cdot \frac{N-n}{N}.$$

This is conservative for an ordered population and anti-conservative under periodicity.

4 Population Total and Proportion

Assuming a randomly ordered population:

$$\hat{\tau} = N\bar{y}_{\text{sy}}, \qquad \widehat{\text{Var}}(\hat{\tau}) = N^2\widehat{\text{Var}}(\bar{y}_{\text{sy}}) = N^2\frac{s^2}{n}\frac{N-n}{N};$$

$$\hat{p}_{\text{sy}} = \bar{y}_{\text{sy}} = \frac{1}{n}\sum y_i \ (y_i \in \{0,1\}), \qquad \widehat{\text{Var}}(\hat{p}_{\text{sy}}) = \frac{\hat{p}_{\text{sy}}\hat{q}_{\text{sy}}}{n-1}\cdot\frac{N-n}{N}.$$

5 Selecting the Sample Size

Under the same assumption the SRS formulas apply: for μ with bound B,

$$n = \frac{N\sigma^2}{(N-1)D+\sigma^2} = \frac{NS^2}{ND+S^2}, \qquad D = \frac{B^2}{4},$$

and for p, $n = \dfrac{Npq}{(N-1)D+pq}$ with $D = B^2/4$.

6 Selected exercises

Motivation. Systematic sampling within strata, the exact gain in an ordered population, and a total under random order.

Exercise 4.1 (Systematic sampling within strata)

N elements in L strata of sizes $N_\ell = n_\ell k$ (common k); independent 1-in-k systematic samples (Method A) in each. With $\bar{y}_\ell$ the stratum sample mean and $\hat{\mu} = \sum_\ell w_\ell \bar{y}_\ell$, $w_\ell = N_\ell/N$, find $\text{E}(\hat{\mu})$ and $\text{Var}(\hat{\mu})$.

Solution. Let $y_{\ell ij}$ be element j of possible sample i in stratum ℓ, $\bar{y}_{\ell i} = \frac{1}{n_\ell}\sum_j y_{\ell ij}$, and μ_ℓ, S_ℓ^2, ρ_ℓ the stratum mean, variance and intraclass correlation.

Step 1. (Unbiasedness.) In stratum ℓ the sample i is chosen with probability $1/k$, so by Case 1 of the unbiasedness subsection, $\text{E}(\bar{y}_\ell) = \frac{1}{k}\sum_i \bar{y}_{\ell i} = \frac{1}{N_\ell}\sum_i\sum_j y_{\ell ij} = \mu_\ell$. Hence $\text{E}(\hat{\mu}) = \sum_\ell w_\ell\mu_\ell = \frac{1}{N}\sum_\ell \tau_\ell = \mu$.

Step 2. (Variance.) The random starts are independent across strata, so the covariances vanish:

$$\text{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \text{Var}(\bar{y}_\ell), \qquad \text{Var}(\bar{y}_\ell) = \frac{1}{k}\sum_{i=1}^{k}(\bar{y}_{\ell i}-\mu_\ell)^2.$$

Step 3. (Intraclass form.) Applying the intraclass-correlation theorem within stratum ℓ,

$$\text{Var}(\hat{\mu}) = \sum_{\ell=1}^{L} w_\ell^2 \frac{S_\ell^2}{n_\ell}\cdot\frac{N_\ell-1}{N_\ell}\left[1+(n_\ell-1)\rho_\ell\right].$$

Step 4. (Interpretation.) $n_\ell = N_\ell/k$: automatic proportional allocation with fraction $1/k$. In random order ($\rho_\ell \approx -1/(N_\ell-1)$) the bracket times $\frac{N_\ell-1}{N_\ell}$ is $1-\frac{1}{k}$, giving proportional stratified sampling. Variances are estimated by the SRS formula per stratum. ■

Exercise 4.2 (Population with a linear trend)

Let $y_t = a + bt$, $t = 1, \dots, N$, with $b \neq 0$ and $N = nk$, $n \geq 2$. Show that

$$\operatorname{Var}(\bar{y}_{\text{sy}}) = \frac{b^2(k^2-1)}{12}, \qquad \operatorname{Var}(\bar{y}_{\text{ran}}) = \frac{b^2(k-1)(N+1)}{12}, \qquad \frac{\operatorname{Var}(\bar{y}_{\text{sy}})}{\operatorname{Var}(\bar{y}_{\text{ran}})} = \frac{k+1}{N+1} < 1.$$

Solution. [Supplement]

Step 1. (A discrete uniform variance.) If I is uniform on $\{1, \dots, k\}$, then $\mathrm{E}I = \frac{k+1}{2}$, $\mathrm{E}I^2 = \frac{(k+1)(2k+1)}{6}$ and

$$\operatorname{Var} I = \frac{(k+1)(2k+1)}{6} - \frac{(k+1)^2}{4} = \frac{(k+1)\big[2(2k+1) - 3(k+1)\big]}{12} = \frac{k^2-1}{12}.$$

Step 2. (Systematic mean.) Sample i consists of $t = i, i+k, \dots, i+(n-1)k$, so $\bar{y}_i = a + b\big(i + \frac{(n-1)k}{2}\big)$. The start I is uniform on $\{1, \dots, k\}$, so $\operatorname{Var}(\bar{y}_{\text{sy}}) = b^2 \operatorname{Var} I = \frac{b^2(k^2-1)}{12}$.

Step 3. (SRS.) Step 1 with k replaced by N gives $\sum_{t=1}^{N} \big(t - \frac{N+1}{2}\big)^2 = N\frac{N^2-1}{12}$, so $S^2 = \frac{b^2}{N-1} \cdot \frac{N(N^2-1)}{12} = \frac{b^2 N(N+1)}{12}$. With $1 - f = 1 - \frac{1}{k}$ and $N = nk$,

$$\operatorname{Var}(\bar{y}_{\text{ran}}) = \frac{k-1}{k} \cdot \frac{1}{n} \cdot \frac{b^2 N(N+1)}{12} = \frac{b^2(k-1)(N+1)}{12}.$$

Step 4. (Ratio.) Dividing, $\frac{(k^2-1)}{(k-1)(N+1)} = \frac{k+1}{N+1}$, which is below 1 because $k < N$ when $n \geq 2$. For $N = 1000$ and $k = 10$ the systematic sample has about 1% of the SRS variance: it cannot pick only small or only large units, which is the "ordered population" case of the text made exact. ■

Exercise 4.3 (Families who rent)

A 1-in-50 systematic sample of tax records ($N = 15{,}200$ households) estimates the number of renting families: of $n = 304$, $\sum y_i = 88$ rent. Estimate the total with a bound, assuming random order with respect to renting.

Solution.

Step 1. $\hat{p}_{\text{sy}} = 88/304 = 0.2895$ and $\hat{\tau} = N\hat{p}_{\text{sy}} = 15{,}200 \times 0.2895 = 4400$.

Step 2. Under the random-order assumption the SRS formulas apply:

$$\widehat{\operatorname{Var}}(\hat{p}_{\text{sy}}) = \frac{\hat{p}\hat{q}}{n-1} \cdot \frac{N-n}{N} = \frac{0.2895 \times 0.7105}{303} \cdot \frac{14{,}896}{15{,}200} = 0.000665.$$

Step 3. $\widehat{\operatorname{Var}}(\hat{\tau}) = 15{,}200^2 \times 0.000665 = 153{,}700$, $2\sqrt{153{,}700} = 784$: 4400 ± 784 families, valid only without periodicity of period 50 related to renting (e.g. apartment blocks of 50). ■

Chapter 5. Ratio Estimation

1 Auxiliary Information

So far $y_i = \mu + \varepsilon_i$, $\mathrm{E}(\varepsilon_i) = 0$. With an auxiliary x related to y on each sampled unit, a better model is

$$y_i = f(x_i) + \varepsilon_i, \qquad \mathrm{E}(y_i) = f(x_i) \quad (\text{e.g. } f(x) = \beta_0 + \beta_1 x),$$

and a known μ_x or τ_x sharpens the estimate of μ_y or τ_y: a line through the origin here (ratio estimation), a general line in Ch. 7 (regression).

Example 5.1: Sugar in a truckload of oranges

Orange prices are based on the total sugar content τ_y of a load; $N\bar{y}$ needs the unknown number of oranges N. Sugar y is closely related to weight x, and the load's total weight τ_x is easy to get. Since

$$\frac{\mu_y}{\mu_x} = \frac{N\mu_y}{N\mu_x} = \frac{\tau_y}{\tau_x} \quad \Longrightarrow \quad \tau_y = \frac{\mu_y}{\mu_x}\tau_x,$$

the **ratio estimator** of the total is

$$\hat{\tau}_y = \frac{\bar{y}}{\bar{x}}\tau_x = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i}\tau_x,$$

which does not require N.

Other ratios: food budget to income; children to residents; dose of a new drug relative to a standard one.

2 Ratio Estimation under SRS

Motivation. For x strongly correlated with y and known τ_x, $\hat{R}\tau_x$ is far more stable than $N\bar{y}$.

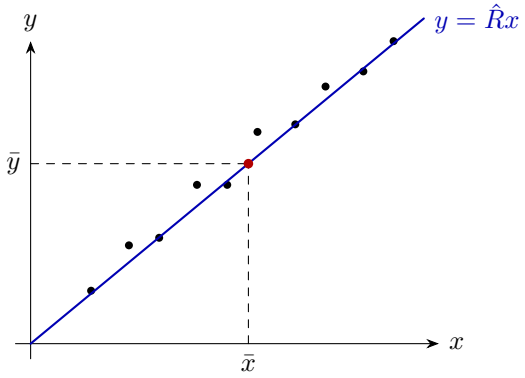


Figure. *A line through the origin with slope $\hat{R} = \bar{y}/\bar{x}$; for $y \approx$ proportional to x, $\hat{\tau}_y = \hat{R}\,\tau_x$ beats $N\bar{y}$.*

Let (x_i, y_i), $i = 1, \ldots, n$, be an SRS of pairs from a population of size N. The population ratio and its estimator are

$$R = \frac{\sum_{i=1}^N y_i}{\sum_{i=1}^N x_i} = \frac{\mu_y}{\mu_x}, \qquad r = \frac{\sum_{i=1}^n y_i}{\sum_{i=1}^n x_i} = \frac{\bar{y}}{\bar{x}}.$$

Theorem 5.2: Ratio estimators

With $f = n/N$ and $s_r^2 = \frac{1}{n-1}\sum_{i=1}^n (y_i - r x_i)^2$,

$$\text{ratio:}\quad r = \bar{y}/\bar{x}, \qquad \widehat{\mathrm{Var}}(r) = \frac{1-f}{n}\cdot\frac{1}{\mu_x^2}\, s_r^2 \quad (\mu_x \text{ replaced by } \bar{x} \text{ if unknown});$$

$$\text{total:}\quad \hat{\tau}_y = r\tau_x, \qquad \widehat{\mathrm{Var}}(\hat{\tau}_y) = \tau_x^2 \widehat{\mathrm{Var}}(r) = \tau_x^2 \frac{1-f}{n}\frac{s_r^2}{\bar{x}^2};$$

$$\text{mean:}\quad \hat{\mu}_y = r\mu_x, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \mu_x^2 \widehat{\mathrm{Var}}(r) = \frac{1-f}{n} s_r^2.$$

The residuals $y_i - r x_i$ measure departure from a line through the origin; if $y_i \approx r x_i$, s_r^2 is small.

Derivation: first-order variance of r

Write $d_i = y_i - R x_i$. For N population units, $\bar{D} = \frac{1}{N}\sum_{i=1}^N (y_i - R x_i) = \mu_y - R\mu_x = 0$. If $\bar{x} \approx \mu_x$,

$$r - R = \frac{\bar{y} - R\bar{x}}{\bar{x}} \approx \frac{\bar{y} - R\bar{x}}{\mu_x} = \frac{1}{\mu_x}\cdot\frac{1}{n}\sum_{i=1}^n (y_i - R x_i) = \frac{\bar{d}}{\mu_x}.$$

Thus $r - R$ is approximately the SRS mean of the d_i divided by μ_x, and by Ch. 2

$$\mathrm{Var}(r) \approx \mathrm{E}(r-R)^2 = \frac{1}{\mu_x^2}\mathrm{E}(\bar{d} - \bar{D})^2 = \frac{1}{\mu_x^2}\cdot\frac{N-n}{N}\cdot\frac{1}{n} S_r^2, \qquad S_r^2 = \frac{1}{N-1}\sum_{i=1}^N (y_i - R x_i)^2.$$

Expanding the square,

$$\sum_{i=1}^N (y_i - R x_i)^2 = \sum_{i=1}^N \left[(y_i - \mu_y) - R(x_i - \mu_x)\right]^2 = (N-1)(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

using $\mu_y = R\mu_x$ and $\sum (x_i - \mu_x)(y_i - \mu_y) = (N-1)\rho S_x S_y$. Hence

$$\mathrm{Var}(r) \approx \frac{1}{\mu_x^2}\cdot\frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y).$$

Replacing R by r and S_r^2 by s_r^2 gives $\widehat{\mathrm{Var}}(r)$. Note that $\mathrm{Var}(r)$ is smallest when $y_i = R x_i$ exactly.

2.1 Correlation form

Expanding s_r^2 gives the equivalent estimate

$$\widehat{\mathrm{Var}}(r) = \frac{1-f}{n}\cdot\frac{1}{\bar{x}^2}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho} s_x s_y), \qquad \hat{\rho} = \frac{s_{xy}}{s_x s_y},$$

with $s_{xy} = \frac{1}{n-1}\sum (x_i - \bar{x})(y_i - \bar{y})$. A larger positive correlation $\hat{\rho}$ means a smaller variance.

Example 5.3: Assessed values of homes

To estimate the relative change in the assessed value of homes over two years (x: earlier value, y: current value), $n = 20$ homes give

$$r = \frac{\sum_{i=1}^{20} y_i}{\sum_{i=1}^{20} x_i} = 1.066, \qquad \sqrt{\widehat{\mathrm{Var}}(r)} = 0.00745, \qquad B = 2\sqrt{\widehat{\mathrm{Var}}(r)} = 0.0149,$$

so an approximate 95% interval for R is [1.0511, 1.0809].

Example 5.4: Oranges, continued

Ten oranges give $\sum y_i = 0.246$ (sugar) and $\sum x_i = 4.350$ (weight); the load weighs $\tau_x = 1800$ pounds:

$$\hat{\tau}_y = \frac{0.246}{4.350}(1800) = 101.79, \qquad 2\sqrt{\widehat{\mathrm{Var}}(\hat{\tau}_y)} = 6.3.$$

3 Bias of the Ratio Estimator

Motivation. The ratio estimator has bias of order $1/n$, negligible relative to its standard error.

r is a ratio of random variables and is generally biased for R. Write $\delta_y = (\bar{y} - \mu_y)/\mu_y$ and $\delta_x = (\bar{x} - \mu_x)/\mu_x$, so

$$r = R\,(1+\delta_y)(1+\delta_x)^{-1} = R(1+\delta_y)(1 - \delta_x + \delta_x^2 - \cdots) \approx R[1 + \delta_y - \delta_x + \delta_x^2 - \delta_x\delta_y].$$

- **Case 1** ($\bar{x} \approx \mu_x$, keep only δ_y): $r \approx R + R\delta_y$, so $\mathrm{E}(r) \approx R$.
- **Case 2** (first order): $\mathrm{E}(r) \approx R + R[\mathrm{E}\delta_y - \mathrm{E}\delta_x] = R$.
- **Case 3** (second order):
$$\mathrm{E}(r) - R \approx R\,\mathrm{E}(\delta_x^2) - R\,\mathrm{E}(\delta_x\delta_y).$$

Theorem 5.5: Approximate bias

$$\mathrm{E}(r) - R \approx \frac{1}{\mu_x^2} \cdot \frac{1-f}{n}(RS_x^2 - \rho S_x S_y).$$

The bias vanishes (to this order) when $RS_x = \rho S_y$, i.e. when $R = \rho S_y/S_x = S_{xy}/S_x^2$ is the least-squares slope.

Proof. Under SRS, $\mathrm{E}[(\bar{x}-\mu_x)^2] = \frac{1-f}{n}S_x^2$ and $\mathrm{E}[(\bar{x}-\mu_x)(\bar{y}-\mu_y)] = \frac{1-f}{n}S_{xy} = \frac{1-f}{n}\rho S_x S_y$. Therefore

$$\mathrm{E}(r) - R \approx \frac{R}{\mu_x^2}\frac{1-f}{n}S_x^2 - \frac{R}{\mu_x\mu_y}\frac{1-f}{n}\rho S_x S_y,$$

and $R/(\mu_x\mu_y) = 1/\mu_x^2$. ∎

Remark 5.6: When is r unbiased?

If $y_i = \beta x_i$ (a line through the origin), then $R = \sum y_i / \sum x_i = \beta$ and also

$$\rho\frac{S_y}{S_x} = \frac{S_{xy}}{S_x^2} = \frac{\frac{1}{N-1}\sum(x_i-\mu_x)\,\beta(x_i-\mu_x)}{\frac{1}{N-1}\sum(x_i-\mu_x)^2} = \beta,$$

so the bias term is zero. The bias is negligible whenever the relation between y and x is a straight line through the origin.

Theorem 5.7: An exact bias bound

$$\mathrm{E}(r) - R = -\frac{\mathrm{Cov}(r,\bar{x})}{\mu_x}, \qquad \frac{|\mathrm{E}(r) - R|}{\sqrt{\mathrm{Var}(r)}} \le \frac{\sqrt{\mathrm{Var}(\bar{x})}}{\mu_x} = CV(\bar{x}).$$

Proof. Since $r\bar{x} = \bar{y}$,

$$\mathrm{Cov}(r,\bar{x}) = \mathrm{E}(r\bar{x}) - \mathrm{E}(r)\mathrm{E}(\bar{x}) = \mu_y - \mathrm{E}(r)\mu_x = \mu_x(R - \mathrm{E}(r)).$$

By $|\mathrm{Cov}(r,\bar{x})| \le \sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$, $|\mathrm{E}(r) - R| \le \frac{1}{\mu_x}\sqrt{\mathrm{Var}(r)}\sqrt{\mathrm{Var}(\bar{x})}$. ■

The bias is small relative to the SE when the CV of $\bar{x}$ is small (moderately large n).

4 Ratio Estimation in Stratified Random Sampling

Motivation. Ratios per stratum (separate) or overall (combined), depending on stratum sizes and ratio differences.

With strata $h = 1, \ldots, L$ (weights $w_h = N_h/N$), there are two ways to build a ratio estimator of μ_y.

Definition 5.8

1. **Separate ratio estimator**: estimate the ratio within each stratum and combine,

$$\hat{\mu}_{yRS} = \sum_{h=1}^{L} w_h\, r_h\, \mu_{xh}, \qquad r_h = \frac{\bar{y}_h}{\bar{x}_h}.$$

   For two strata A and B: $\hat{\mu}_{yRS} = \frac{N_A}{N}\frac{\bar{y}_A}{\bar{x}_A}\mu_{xA} + \frac{N_B}{N}\frac{\bar{y}_B}{\bar{x}_B}\mu_{xB}$.

2. **Combined ratio estimator**: estimate μ_y and μ_x by the stratified means and take one ratio,

$$\hat{\mu}_{yRC} = r_C\, \mu_x, \qquad r_C = \frac{\bar{y}_{\mathrm{st}}}{\bar{x}_{\mathrm{st}}}, \quad \bar{y}_{\mathrm{st}} = \sum_h w_h \bar{y}_h,\ \bar{x}_{\mathrm{st}} = \sum_h w_h \bar{x}_h.$$

Theorem 5.9: Estimated variances

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRS}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h} \cdot \frac{\sum_{i=1}^{n_h}(y_{hi} - r_h x_{hi})^2}{n_h - 1},$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yRC}) = \sum_{h=1}^{L} w_h^2 \frac{N_h - n_h}{N_h n_h}\, s_{rh}^2,$$

where s_{rh}^2 is the sample variance of the terms $y_{hi} - r_C x_{hi}$ in stratum h.

Derivation: which one is better?

The population versions are $\mathrm{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,S}^2$ and $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h} S_{h,C}^2$, with $R_h = \mu_{yh}/\mu_{xh}$, $R_C = \mu_y/\mu_x$ and

$$(N_h - 1)S_{h,S}^2 = \sum_i (y_{hi} - R_h x_{hi})^2 = (N_h - 1)(S_{yh}^2 + R_h^2 S_{xh}^2 - 2R_h \rho_h S_{xh} S_{yh}),$$

$$(N_h-1)S_{h,C}^2 = \sum_i \left[(y_{hi}-\mu_{yh}) - R_C(x_{hi}-\mu_{xh})\right]^2$$
$$= (N_h-1)(S_{yh}^2 + R_C^2 S_{xh}^2 - 2R_C\rho_h S_{xh}S_{yh}).$$

(The combined deviation $\hat{\mu}_{yRC} - \mu_y \approx \bar{y}_{\text{st}} - R_C\bar{x}_{\text{st}} = \bar{d}_{\text{st}}$ is a stratified mean of $d_{hi} = y_{hi} - R_C x_{hi}$, whose stratum means are $\mu_{yh} - R_C\mu_{xh}$.) Subtracting,

$$S_{h,C}^2 - S_{h,S}^2 = (R_C^2 - R_h^2)S_{xh}^2 - 2(R_C - R_h)\rho_h S_{xh}S_{yh}$$
$$= (R_C - R_h)^2 S_{xh}^2 + 2(R_C - R_h)(R_h S_{xh}^2 - \rho_h S_{xh}S_{yh}).$$

The last bracket is zero when each stratum ratio is (approximately) unbiased, $R_h S_{xh} = \rho_h S_{yh}$. Then

$$\text{Var}(\hat{\mu}_{yRC}) - \text{Var}(\hat{\mu}_{yRS}) \approx \sum_h w_h^2 \frac{1-f_h}{n_h}(R_C - R_h)^2 S_{xh}^2 \ \geq 0.$$

Remark 5.10: Practical guidance

- When the stratum ratios R_h differ greatly, the separate estimator is more efficient.
- But the stratum ratio biases add up: badly biased for small n_h, negligible for n_h over about 50. With small strata use the combined estimator.

5 Ratio Estimator versus the Sample Mean

Motivation. Ratio beats $\bar{y}$ when ρ exceeds half the ratio of the coefficients of variation.

Compare $\bar{y}$ and $\hat{\mu}_y = r\mu_x$ by relative efficiency $\text{RE}(\hat{\theta}_1, \hat{\theta}_2) = \text{Var}(\hat{\theta}_1)/\text{Var}(\hat{\theta}_2)$:

$$\text{Var}(\bar{y}) = \frac{1-f}{n}S_y^2, \qquad \text{Var}(\hat{\mu}_y) \approx \frac{1-f}{n}(S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y),$$

$$\text{RE}(\bar{y}, \hat{\mu}_y) = \frac{\text{Var}(\bar{y})}{\text{Var}(\hat{\mu}_y)} = \frac{S_y^2}{S_y^2 + R^2 S_x^2 - 2R\rho S_x S_y}.$$

Theorem 5.11: When the ratio estimator wins

$\text{RE}(\bar{y}, \hat{\mu}_y) > 1$ if and only if

$$\rho > \frac{1}{2}\cdot\frac{S_x/\mu_x}{S_y/\mu_y} = \frac{1}{2}\cdot\frac{CV_x}{CV_y}.$$

In particular, if x and y have equal coefficients of variation, the ratio estimator is better as soon as $\rho > 1/2$.

Proof. $\text{RE} > 1 \iff R^2 S_x^2 < 2R\rho S_x S_y \iff RS_x < 2\rho S_y \iff \rho > \frac{RS_x}{2S_y}$, and $R = \mu_y/\mu_x$. The sample version replaces everything by estimates: $\hat{\rho} > \frac{1}{2}\frac{s_x/\bar{x}}{s_y/\bar{y}}$. ■

Efficient for highly positively correlated x, y; the estimated relative efficiency is meaningful only for large samples.

6 Selected exercises

Motivation. The ratio estimator is model-optimal, and a computation for a total.

Exercise 5.1 (The ratio estimator as a weighted least-squares fit)

Consider the model $y_i = \beta x_i + \varepsilon_i$, $i = 1, \dots, n$, with fixed $x_i > 0$, $\mathrm{E}(\varepsilon_i) = 0$, $\mathrm{Var}(\varepsilon_i) = \sigma^2 x_i$ and uncorrelated errors.
(a) Show that $r = \sum y_i / \sum x_i$ minimizes the weighted sum of squares $\sum_i (y_i - \beta x_i)^2 / x_i$.
(b) Show that r is unbiased for β with $\mathrm{Var}(r) = \sigma^2 / \sum_i x_i$, and that it has the smallest variance among all linear unbiased estimators $\sum_i a_i y_i$ of β.

Solution. [Supplement]

(a) $Q(\beta) = \sum_i (y_i - \beta x_i)^2 / x_i$ is a quadratic in β with $Q'(\beta) = -2\sum_i (y_i - \beta x_i) = -2(\sum_i y_i - \beta \sum_i x_i)$ and $Q'' = 2\sum_i x_i > 0$, so the unique minimizer is $\beta = \sum y_i / \sum x_i = r$.

(b)

Step 1. (Unbiasedness and variance.) $r = \sum_i a_i^0 y_i$ with $a_i^0 = 1/\sum_j x_j$. Then $\mathrm{E}(r) = \beta \sum_i a_i^0 x_i = \beta$ and $\mathrm{Var}(r) = \sigma^2 \sum_i (a_i^0)^2 x_i = \sigma^2 / \sum_i x_i$.

Step 2. (Unbiasedness constraint.) A linear estimator $\sum_i a_i y_i$ has mean $\beta \sum_i a_i x_i$, so it is unbiased for every β iff $\sum_i a_i x_i = 1$. Its variance is $\sigma^2 \sum_i a_i^2 x_i$.

Step 3. (Cauchy–Schwarz.) Writing $a_i x_i = (a_i \sqrt{x_i})\sqrt{x_i}$,

$$1 = \Big(\sum_i a_i x_i\Big)^2 \le \Big(\sum_i a_i^2 x_i\Big)\Big(\sum_i x_i\Big),$$

so $\sigma^2 \sum_i a_i^2 x_i \ge \sigma^2 / \sum_i x_i = \mathrm{Var}(r)$, with equality iff $a_i \sqrt{x_i} \propto \sqrt{x_i}$, i.e. a_i constant, i.e. $a_i = a_i^0$.

So it is BLUE when $y \propto x$ with error variance $\propto x$, as for the oranges. ■

Exercise 5.2 (Sales of television sets)

Total TV sales τ_y this quarter: an SRS of $n = 13$ of $N = 123$ offices gives previous-year x_i and current y_i, with previous-year total $\tau_x = 128{,}200$.

x_i	550	720	1500	1020	620	980	928	1200	1350	1750	670	729	1530
y_i	610	780	1600	1030	600	1050	977	1440	1570	2210	980	865	1710

Estimate τ_y by the ratio estimator and place a bound on the error.

Solution.

Step 1. $\sum x_i = 13{,}547$, $\sum y_i = 15{,}422$, so $r = 15{,}422/13{,}547 = 1.1384$ and $\hat\tau_y = r\tau_x = 1.1384 \times 128{,}200 = 145{,}944$.

Step 2. The residual variance is $s_r^2 = \frac{1}{12}\sum_i (y_i - r x_i)^2 = 12{,}990$, and $\mu_x = \tau_x / N = 1042.3$ is known.

Step 3. By the ratio-estimator theorem,

$$\widehat{\mathrm{Var}}(\hat\tau_y) = \tau_x^2 \frac{1-f}{n}\frac{s_r^2}{\mu_x^2} = N^2 \frac{1-f}{n} s_r^2 = 123^2 \cdot \frac{1 - 13/123}{13} \cdot 12{,}990 = 1.352 \times 10^7,$$

so the bound is $2\sqrt{1.352 \times 10^7} = 7354$ and τ_y is estimated by $145{,}944 \pm 7354$.

Step 4. For comparison, $N\bar y = 123 \times 1186.3 = 145{,}916$ has $s_y^2 = 231{,}543$ and bound

$$2N\sqrt{(1-f)s_y^2/n} \approx 31{,}050 :$$

the ratio estimator is about four times as precise, since current and previous sales are almost proportional. ■

Chapter 6. Simple Linear Regression

Simple linear regression, the basis of the regression estimator (Ch. 7), compresses n-dimensional data into $(\hat\beta_0, \hat\beta_1)$: data = model structure + error.

1 The Model

Definition 6.1

The simple linear regression model with a single regressor x is

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \dots, n,$$

with uncorrelated errors of mean 0 and variance σ^2; x is fixed, y random; β_0, β_1 are the **regression coefficients**.

Then

$$\mathrm{E}(y \mid x) = \beta_0 + \beta_1 x = f(x), \qquad \mathrm{Var}(y \mid x) = \mathrm{Var}(\varepsilon) = \sigma^2,$$

so the true regression line is the line of mean responses. The goals are to estimate β_0, β_1 and to test $H_0 : \beta_1 = 0$.

2 Least-Squares Estimation

Motivation. With an intercept, a regression line replaces the ratio model; we review least squares.

The least-squares (LS) estimators minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2 = \sum_{i=1}^{n} \varepsilon_i^2.$$

Setting $\partial S/\partial \beta_0 = \partial S/\partial \beta_1 = 0$ gives the **normal equations**

$$n\hat\beta_0 + \hat\beta_1 \sum x_i = \sum y_i, \qquad \hat\beta_0 \sum x_i + \hat\beta_1 \sum x_i^2 = \sum x_i y_i.$$

Theorem 6.2: LS estimators

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \frac{\sum (x_i - \bar x)(y_i - \bar y)}{\sum (x_i - \bar x)^2} = \frac{\sum (x_i - \bar x) y_i}{\sum (x_i - \bar x)^2}, \qquad \hat\beta_0 = \bar y - \hat\beta_1 \bar x.$$

The **fitted values** are $\hat y_i = \hat\beta_0 + \hat\beta_1 x_i = \widehat{\mathrm{E}}(y_i \mid x_i)$ and the **residuals** are $e_i = y_i - \hat y_i$, so $y_i = \hat\beta_0 + \hat\beta_1 x_i + e_i$; residuals are the basis of model-adequacy checks (residual analysis).

Example 6.3: Rocket propellant data, $n = 20$

y = shear strength, x = age of propellant. From $\bar x = 13.3625$, $\bar y = 2131.358$, $S_{xx} = 1106.559$, $S_{xy} = -41112.65$:

$$\hat\beta_1 = \frac{-41112.65}{1106.559} = -37.15, \qquad \hat\beta_0 = 2131.358 + 37.15 \times 13.3625 = 2627.82,$$

so $\hat y = 2627.82 - 37.15x$. R output: intercept 2627.822 (SE 44.184), slope -37.154 (SE 2.889, $t = -12.86$, $p = 1.64 \times 10^{-10}$).

2.1 Properties of the estimators

Theorem 6.4

$\mathrm{E}(\hat\beta_1) = \beta_1$, $\mathrm{E}(\hat\beta_0) = \beta_0$, and

$$\mathrm{Var}(\hat\beta_1) = \frac{\sigma^2}{S_{xx}}, \qquad \mathrm{Var}(\hat\beta_0) = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \mathrm{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x\frac{\sigma^2}{S_{xx}}.$$

Derivation: proof

Write $\hat\beta_1 = \sum c_i y_i$ with $c_i = (x_i - \bar x)/S_{xx}$. Then $\sum c_i = 0$ and $\sum c_i x_i = \sum (x_i - \bar x)x_i/S_{xx} = 1$, so

$$\mathrm{E}(\hat\beta_1) = \sum c_i(\beta_0 + \beta_1 x_i) = \beta_0 \sum c_i + \beta_1 \sum c_i x_i = \beta_1,$$

$$\mathrm{Var}(\hat\beta_1) = \sigma^2 \sum c_i^2 = \sigma^2 \frac{S_{xx}}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}}.$$

Since $\mathrm{E}(\bar y) = \beta_0 + \beta_1 \bar x$, $\mathrm{E}(\hat\beta_0) = \beta_0 + \beta_1 \bar x - \bar x \beta_1 = \beta_0$. With $\bar y = \sum d_i y_i$, $d_i = 1/n$,

$$\mathrm{Cov}(\bar y, \hat\beta_1) = \sigma^2 \sum d_i c_i = \frac{\sigma^2}{n} \sum c_i = 0,$$

hence

$$\mathrm{Var}(\hat\beta_0) = \mathrm{Var}(\bar y) + \bar x^2 \mathrm{Var}(\hat\beta_1) - 2\bar x \mathrm{Cov}(\bar y, \hat\beta_1) = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big),$$

$$\mathrm{Cov}(\hat\beta_0, \hat\beta_1) = \mathrm{Cov}(\bar y, \hat\beta_1) - \bar x \mathrm{Var}(\hat\beta_1) = -\bar x \sigma^2 / S_{xx}.$$

2.2 Properties of the fit

Theorem 6.5

(1) $\sum e_i = 0$, equivalently $\sum y_i = \sum \hat y_i$;

(2) the fitted line passes through $(\bar x, \bar y)$;

(3) $\sum x_i e_i = 0$;

(4) $\sum \hat y_i e_i = 0$.

Proof. (1) and (3) are the two normal equations: $\sum (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$ and $\sum x_i (y_i - \hat\beta_0 - \hat\beta_1 x_i) = 0$. (2): at $x = \bar x$, $\hat y = \hat\beta_0 + \hat\beta_1 \bar x = \bar y$. (4): $\sum \hat y_i e_i = \hat\beta_0 \sum e_i + \hat\beta_1 \sum x_i e_i = 0$. ■

2.3 Centered model

Rewriting

$$y_i = (\beta_0 + \beta_1 \bar x) + \beta_1 (x_i - \bar x) + \varepsilon_i = \beta_0' + \beta_1 (x_i - \bar x) + \varepsilon_i$$

gives $\hat\beta_0' = \bar y$ and $\hat\beta_1 = S_{xy}/S_{xx}$, and now $\mathrm{Cov}(\hat\beta_0', \hat\beta_1) = \mathrm{Cov}(\bar y, \hat\beta_1) = 0$: centering makes the estimators uncorrelated.

3 Estimation of σ^2

Motivation. Inference on the slope needs the residual mean square.

Theorem 6.6: Sum-of-squares decomposition

$$\underbrace{\sum (y_i-\bar y)^2}_{SS_T}=\underbrace{\sum (y_i-\hat y_i)^2}_{SS_E}+\underbrace{\sum (\hat y_i-\bar y)^2}_{SS_R},\qquad SS_E=SS_T-\hat\beta_1 S_{xy},\quad SS_R=\hat\beta_1 S_{xy}.$$

Proof. Expanding $\sum (y_i-\hat y_i+\hat y_i-\bar y)^2$ leaves the cross term $2\sum e_i(\hat y_i-\bar y)=2(\sum e_i\hat y_i-\bar y\sum e_i)=0$. Since $e_i=(y_i-\bar y)-\hat\beta_1(x_i-\bar x)$,

$$SS_E=SS_T+\hat\beta_1^2 S_{xx}-2\hat\beta_1 S_{xy}=SS_T-\hat\beta_1 S_{xy}.$$

■

SS_E has $n-2$ degrees of freedom, and the unbiased estimator of σ^2 is

$$\hat\sigma^2=\frac{SS_E}{n-2}=MS_E\quad\text{(the mean square error)}.$$

$\hat\sigma$ is the standard error of regression; it is a model-dependent estimate of σ.

Example 6.7: Rocket propellant

SS_T = 1,693,737.6 and SS_E = 1,693,737.6 − (−37.15)(−41112.65) = 166,402.65, so $\hat\sigma^2$ = 166402.65/18 = 9244.59.

4 Hypothesis Testing on the Slope

Motivation. A non-significant slope means the auxiliary variable helps little.

Now assume $\varepsilon_i \overset{\text{iid}}{\sim} N(0,\sigma^2)$. Then $\hat\beta_1 \sim N(\beta_1,\sigma^2/S_{xx})$.

Theorem 6.8: t-test for significance of regression

For $H_0:\beta_1=0$ versus $H_1:\beta_1\neq 0$,

$$t_0=\frac{\hat\beta_1}{\operatorname{se}(\hat\beta_1)},\qquad \operatorname{se}(\hat\beta_1)=\sqrt{MS_E/S_{xx}},$$

which is t_{n-2} under H_0. Reject H_0 if $|t_0|>t_{\alpha/2,n-2}$ (equivalently, if the P-value $\le\alpha$). If σ^2 were known, $Z_0=\hat\beta_1/\sqrt{\sigma^2/S_{xx}}\sim N(0,1)$ would be used instead.

Failing to reject H_0 means there is no evidence of a *linear* relationship between x and y.

Example 6.9: Rocket propellant

$t_0=-37.15/\sqrt{9244.59/1106.56}=-12.85$ and $t_{0.025,18}=2.101$, so H_0 is rejected; the P-value is 1.64×10^{-10}.

4.1 Analysis of variance

Theorem 6.10: F-test

With degrees of freedom $(n-1) = 1 + (n-2)$,

$$F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{MS_R}{MS_E} \sim F_{1,n-2} \quad \text{under } H_0,$$

and H_0 is rejected if $F_0 > F_{\alpha,1,n-2}$. Since $\mathrm{E}(MS_R) = \sigma^2 + \beta_1^2 S_{xx}$ and $\mathrm{E}(MS_E) = \sigma^2$, F_0 is near 1 under H_0 and large otherwise. In simple regression $F_0 = t_0^2$.

ANOVA gives the same answer as the t-test here, but it is the natural tool for multiple regression.

Example 6.11: Rocket propellant

$SS_R = (-37.15)(-41112.65) = 1{,}527{,}334.95$, $SS_E = 166{,}402.65$,

$$F_0 = \frac{1527334.95/1}{166402.65/18} = 165.21 > F_{0.05,1,18} = 4.41,$$

so H_0 is rejected (R: $F = 165.4$ on 1 and 18 df, $p = 1.64 \times 10^{-10}$).

5 Interval Estimation

5.1 Coefficients

Under normal errors, $(\hat\beta_j - \beta_j)/\mathrm{se}(\hat\beta_j) \sim t_{n-2}$, so

$$\hat\beta_1 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_1), \qquad \hat\beta_0 \pm t_{\alpha/2,n-2}\,\mathrm{se}(\hat\beta_0),$$

$$\mathrm{se}(\hat\beta_1) = \sqrt{\frac{MS_E}{S_{xx}}}, \qquad \mathrm{se}(\hat\beta_0) = \sqrt{MS_E\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

Example 6.12

For the rocket data with $t_{0.025,18} = 2.101$:

$$2534.99 \le \beta_0 \le 2720.65, \qquad -43.22 \le \beta_1 \le -31.08.$$

5.2 Error variance

Since $(n-2)MS_E/\sigma^2 \sim \chi^2_{n-2}$,

$$\frac{(n-2)MS_E}{\chi^2_{\alpha/2,n-2}} \le \sigma^2 \le \frac{(n-2)MS_E}{\chi^2_{1-\alpha/2,n-2}}.$$

5.3 Mean response

For x_0 within the range of the data, the estimate is $\hat\mu_{y|x_0} = \hat\beta_0 + \hat\beta_1 x_0$, with $\mathrm{E}(\hat\mu_{y|x_0}) = \beta_0 + \beta_1 x_0$ and

$$\mathrm{Var}(\hat\mu_{y|x_0}) = \mathrm{Var}(\hat\beta_0) + x_0^2\mathrm{Var}(\hat\beta_1) + 2x_0\mathrm{Cov}(\hat\beta_0, \hat\beta_1)$$

$$= \sigma^2\Big(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\Big) + x_0^2\frac{\sigma^2}{S_{xx}} - 2x_0\bar{x}\frac{\sigma^2}{S_{xx}} = \sigma^2\Big(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\Big).$$

Hence

$$\hat{\mu}_{y|x_0} \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\Big)}.$$

At $x_0 = \bar{x} = 13.3625$ in the rocket data the 95% interval is $2086.2 \le \mathrm{E}(y \mid x_0) \le 2176.6$. The interval is narrowest at $\bar{x}$ and widens away from it.

5.4 Prediction of a new observation

The point prediction of y_0 at x_0 is $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$. Since the new y_0 is independent of the data,

$$\mathrm{E}(\hat{y}_0 - y_0)^2 = \mathrm{E}[\hat{y}_0 - \mathrm{E}(y_0)]^2 + \mathrm{E}[y_0 - \mathrm{E}(y_0)]^2 = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\Big),$$

and the $100(1-\alpha)\%$ prediction interval is

$$\hat{y}_0 \pm t_{\alpha/2,n-2}\sqrt{MS_E\Big(1 + \frac{1}{n} + \frac{(x_0-\bar{x})^2}{S_{xx}}\Big)}.$$

For the rocket data at $x_0 = 10$: $[2048.32,\ 2464.32]$.

6 Coefficient of Determination

Definition 6.13

$R^2 = \dfrac{SS_R}{SS_T} = 1 - \dfrac{SS_E}{SS_T}$, $0 \le R^2 \le 1$: the proportion of variation in y explained by x.

In simple regression,

$$R^2 = \frac{\hat{\beta}_1 S_{xy}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2, \qquad F_0 = \frac{SS_R/1}{SS_E/(n-2)} = \frac{R^2(n-2)}{1-R^2}.$$

For the rocket data $R^2 = 1527334.95/1693737.6 = 0.9018 = (-0.9497)^2$: 90.18% of the variability in strength is explained by age.

7 Selected exercises

Motivation. Proofs of the unbiasedness of MS_E, the expected regression mean square and LS optimality, and the no-intercept model linking to Ch. 5.

Exercise 6.1 (Expected mean squares)

In the simple linear regression model show that
(a) $\mathrm{E}(SS_R) = \sigma^2 + \beta_1^2 S_{xx}$;
(b) $\mathrm{E}(SS_E) = (n-2)\sigma^2$, so MS_E is unbiased for σ^2;
(c) $F_0 = t_0^2$.

Solution. [Supplement]

(a) Since $SS_R = \hat\beta_1 S_{xy}$ and $S_{xy} = \hat\beta_1 S_{xx}$, $SS_R = \hat\beta_1^2 S_{xx}$. Hence, by the theorem on the properties of the estimators,

$$\mathrm{E}(SS_R) = S_{xx}\big[\mathrm{Var}(\hat\beta_1) + (\mathrm{E}\hat\beta_1)^2\big] = S_{xx}\Big(\frac{\sigma^2}{S_{xx}} + \beta_1^2\Big) = \sigma^2 + \beta_1^2 S_{xx}.$$

(b)

Step 1. Averaging the model, $\bar y = \beta_0 + \beta_1\bar x + \bar\varepsilon$, so $y_i - \bar y = \beta_1(x_i - \bar x) + (\varepsilon_i - \bar\varepsilon)$.

Step 2. Squaring and summing, the cross term $2\beta_1\sum(x_i-\bar x)(\varepsilon_i-\bar\varepsilon)$ has mean zero, and

$$\mathrm{E}\sum(\varepsilon_i-\bar\varepsilon)^2 = \mathrm{E}\Big[\sum\varepsilon_i^2 - n\bar\varepsilon^2\Big] = n\sigma^2 - n\cdot\frac{\sigma^2}{n} = (n-1)\sigma^2,$$

using that the errors are uncorrelated. Hence $\mathrm{E}(SS_T) = \beta_1^2 S_{xx} + (n-1)\sigma^2$.

Step 3. By the sum-of-squares decomposition and (a), $\mathrm{E}(SS_E) = \mathrm{E}(SS_T) - \mathrm{E}(SS_R) = (n-1)\sigma^2 - \sigma^2 = (n-2)\sigma^2$.

(c) $F_0 = \dfrac{\hat\beta_1^2 S_{xx}}{MS_E} = \Big(\dfrac{\hat\beta_1}{\sqrt{MS_E/S_{xx}}}\Big)^2 = t_0^2$. Under H_0, (a) and (b) give $\mathrm{E}(MS_R) = \mathrm{E}(MS_E) = \sigma^2$, which is why F_0 is near 1. ■

Exercise 6.2 (Least squares is best among linear unbiased estimators)

Show that $\hat\beta_1$ has the smallest variance among all estimators of the form $\sum_i a_i y_i$ that are unbiased for β_1 whatever β_0, β_1 are.

Solution. [Supplement]

Step 1. (Constraints.) $\mathrm{E}\sum a_i y_i = \beta_0\sum a_i + \beta_1\sum a_i x_i$ equals β_1 for all (β_0,β_1) iff $\sum a_i = 0$ and $\sum a_i x_i = 1$. Its variance is $\sigma^2\sum a_i^2$.

Step 2. (Decomposition.) Let $c_i = (x_i - \bar x)/S_{xx}$, so $\hat\beta_1 = \sum c_i y_i$ and the c_i satisfy both constraints. Put $d_i = a_i - c_i$; then $\sum d_i = 0$ and $\sum d_i x_i = 0$.

Step 3. (Orthogonality.) $\sum c_i d_i = \frac{1}{S_{xx}}\big(\sum d_i x_i - \bar x\sum d_i\big) = 0$, so

$$\mathrm{Var}\Big(\sum a_i y_i\Big) = \sigma^2\sum(c_i+d_i)^2 = \sigma^2\sum c_i^2 + \sigma^2\sum d_i^2 \;\ge\; \sigma^2\sum c_i^2 = \mathrm{Var}(\hat\beta_1),$$

with equality iff all $d_i = 0$, i.e. the estimator is $\hat\beta_1$ itself. This is the Gauss–Markov theorem for the slope; normality is not needed. ■

Exercise 6.3 (Regression through the origin)

Let $y_i = \beta x_i + \varepsilon_i$, $i = 1,\dots,n$, with fixed x_i not all zero and uncorrelated errors of mean 0 and variance σ^2.
(a) Find the LS estimator $\hat\beta$, its mean and variance.
(b) With $e_i = y_i - \hat\beta x_i$, show that $\sum x_i e_i = 0$ but in general $\sum e_i \neq 0$.
(c) Show that $\mathrm{E}\sum e_i^2 = (n-1)\sigma^2$, so $\hat\sigma^2 = \sum e_i^2/(n-1)$ is unbiased.

Solution. [Supplement]

(a) $\frac{d}{d\beta}\sum(y_i-\beta x_i)^2 = -2\sum x_i(y_i-\beta x_i) = 0$ gives $\hat\beta = \sum x_i y_i/\sum x_i^2$, the unique minimizer since the second derivative $2\sum x_i^2$ is positive. Then $\mathrm{E}\hat\beta = \beta\sum x_i^2/\sum x_i^2 = \beta$ and $\mathrm{Var}\hat\beta = \sigma^2\sum x_i^2/(\sum x_i^2)^2 = \sigma^2/\sum x_i^2$.

(b) The normal equation of (a) is $\sum x_i e_i = 0$. There is no intercept, hence no equation $\sum e_i = 0$: for the data $(x, y) = (1, 1), (2, 1)$, $\hat{\beta} = 3/5$ and $e_1 + e_2 = 0.4 - 0.2 = 0.2 \neq 0$.

(c)

Step 1. By (b), $\sum e_i^2 = \sum e_i y_i - \hat{\beta} \sum e_i x_i = \sum e_i y_i = \sum y_i^2 - \hat{\beta} \sum x_i y_i = \sum y_i^2 - \hat{\beta}^2 \sum x_i^2$.

Step 2. $\mathrm{E} \sum y_i^2 = \sum (\sigma^2 + \beta^2 x_i^2) = n\sigma^2 + \beta^2 \sum x_i^2$, and by (a) $\mathrm{E}(\hat{\beta}^2) \sum x_i^2 = (\sigma^2 / \sum x_i^2 + \beta^2) \sum x_i^2 = \sigma^2 + \beta^2 \sum x_i^2$.

Step 3. Subtracting, $\mathrm{E} \sum e_i^2 = (n-1)\sigma^2$: only one degree of freedom is lost, for the single coefficient.

With constant variance the no-intercept slope is $\sum x_i y_i / \sum x_i^2$; with variance $\propto x_i$ it is $\sum y_i / \sum x_i$ (Exercise 5.1): the ratio estimator is regression through the origin. ■

Chapter 7. Regression Estimation

1 The Regression Estimator

Ratio estimation suits a line through the origin; with a nonzero intercept use the regression line. As in Ch. 6, x is fixed (observed) and y random:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i = \beta_0' + \beta_1(x_i - \bar{x}) + \varepsilon_i, \qquad \hat{\beta}_0' = \bar{y}, \quad \hat{\beta}_1 = b = \frac{\sum(y_i - \bar{y})(x_i - \bar{x})}{\sum(x_i - \bar{x})^2},$$

so the fitted mean response is $\widehat{\mathrm{E}}(y \mid x) = \bar{y} + b(x - \bar{x})$. Evaluating it at the known population mean μ_x gives the estimator.

Definition 7.1

The **regression estimator** of μ_y is

$$\hat{\mu}_{yL} = \widehat{\mathrm{E}}(y \mid \mu_x) = \bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_{i=1}^{n}\left[y_i - b(x_i - \mu_x)\right],$$

i.e. the sample mean of the adjusted values $y_i - b(x_i - \mu_x)$.

If the sample happens to have $\bar{x} < \mu_x$ and $b > 0$, the sample mean $\bar{y}$ is probably too small as well, and $\hat{\mu}_{yL}$ corrects it upward by $b(\mu_x - \bar{x})$.

Theorem 7.2: Model unbiasedness

Under the model $\mathrm{E}(y_i) = \beta_0 + \beta_1 x_i$ with fixed x's,

$$\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1 \mu_x = \mathrm{E}(y \mid \mu_x).$$

Proof. $\mathrm{E}(\bar{y}) = \beta_0 + \beta_1 \bar{x}$ and $\mathrm{E}(b) = \beta_1$ (Ch. 6), so $\mathrm{E}(\hat{\mu}_{yL}) = \beta_0 + \beta_1\bar{x} + \beta_1(\mu_x - \bar{x}) = \beta_0 + \beta_1\mu_x$. ■

2 Variance of the Regression Estimator

2.1 Fixed slope

First take a fixed constant β and $\bar{y}_{\mathrm{reg}} = \bar{y} + \beta(\mu_x - \bar{x})$. This is the SRS mean of $u_i = y_i - \beta(x_i - \mu_x)$, whose population mean is μ_y, so it is exactly unbiased.

Theorem 7.3

$$\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}(S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2),$$

which is minimized at $\beta = S_{xy}/S_x^2$ (the population LS slope), where

$$\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}\left(S_y^2 - \frac{S_{xy}^2}{S_x^2}\right) = \frac{1-f}{n} S_y^2(1-\rho^2).$$

Proof. By Ch. 2, $\mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n} S_u^2$ with

$$S_u^2 = \frac{1}{N-1}\sum_{i=1}^{N}\left[(y_i - \mu_y) - \beta(x_i - \mu_x)\right]^2$$

$$= \frac{1}{N-1}\Big[\sum (y_i-\mu_y)^2 - 2\beta\sum (x_i-\mu_x)(y_i-\mu_y) + \beta^2\sum (x_i-\mu_x)^2\Big] = S_y^2 - 2\beta S_{xy} + \beta^2 S_x^2.$$

This quadratic in β is minimized at $\beta = S_{xy}/S_x^2$, giving $S_y^2 - S_{xy}^2/S_x^2 = S_y^2(1-\rho^2)$. ■

2.2 Estimated slope

With b estimated, write $\xi_i = (y_i-\mu_y) - \beta(x_i-\mu_x)$ for $\beta = S_{xy}/S_x^2$, so $y_i = \mu_y + \beta(x_i-\mu_x) + \xi_i$. Then

$$b = \frac{\sum (x_i-\bar{x})y_i}{\sum (x_i-\bar{x})^2} = \beta\frac{\sum (x_i-\bar{x})(x_i-\mu_x)}{\sum (x_i-\bar{x})^2} + \frac{\sum (x_i-\bar{x})\xi_i}{\sum (x_i-\bar{x})^2} = \beta + \frac{\sum (x_i-\bar{x})\xi_i}{\sum (x_i-\bar{x})^2},$$

and the last term is of smaller order since the ξ_i are uncorrelated with x in the population. Hence $b \approx \beta$ in large samples and

$$\mathrm{Var}(\hat{\mu}_{yL}) \approx \mathrm{Var}(\bar{y}_{\mathrm{reg}}) = \frac{1-f}{n}S^2, \qquad S^2 = \frac{1}{N-1}\sum_{i=1}^{N}\big[(y_i-\mu_y) - \beta(x_i-\mu_x)\big]^2.$$

Theorem 7.4: Estimated variance

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{N-n}{Nn}\,MS_E, \qquad MS_E = \frac{1}{n-2}\sum_{i=1}^{n}\big[(y_i-\bar{y}) - b(x_i-\bar{x})\big]^2,$$

the mean square error from the simple linear regression of y on x.

The sample analogue of S^2 is $\frac{1}{n-1}\sum e_i^2$ with residuals $e_i = (y_i-\bar{y}) - b(x_i-\bar{x})$; dividing by $n-2$ instead (two estimated coefficients) gives MS_E, and the difference is negligible for moderate n.

Example 7.5: Calculus marks

A mathematics achievement test was taken by all $N = 486$ entering students, with mean $\mu_x = 52$. For $n = 10$ sampled students:

Student	1	2	3	4	5	6	7	8	9	10
Test (x)	39	43	21	64	57	47	28	75	34	52
Final (y)	65	78	52	82	92	89	73	98	56	75

Here $\bar{x} = 46$, $\bar{y} = 76$, and the R fit gives $\hat{y} = 40.78 + 0.7656x$ with residual standard error 8.704 on 8 df ($MS_E = 75.76$, $R^2 = 0.705$). Then

$$\hat{\mu}_{yL} = 76 + 0.7656(52-46) = 80.6,$$

$$\widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{486-10}{486\times 10}(75.76) = 7.42, \qquad B = 2\sqrt{7.42} = 5.45.$$

The fitted intercept (40.8) is far from zero, so a ratio estimator would be inappropriate here: $r\mu_x = (760/460)(52) = 85.9$ overshoots.

3 Relative Efficiency: Mean, Ratio and Regression

$\mathrm{RE}(E_1/E_2) = \mathrm{Var}(E_2)/\mathrm{Var}(E_1)$; above 1, E_1 is preferred (estimated versions for large samples). The estimated variances share $\frac{1-f}{n}$:

$$\widehat{\mathrm{Var}}(\bar{y}) = \frac{1-f}{n}s_y^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_y) = \frac{1-f}{n}(s_y^2 + r^2 s_x^2 - 2r\hat{\rho}s_x s_y), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yL}) = \frac{1-f}{n}s_y^2(1-\hat{\rho}^2).$$

Derivation: the regression variance in correlation form

$$\frac{1}{n-1}\sum\left[(y_i-\bar{y})-b(x_i-\bar{x})\right]^2 = \frac{1}{n-1}\left[S_{yy}-2b\,S_{xy}+b^2S_{xx}\right]$$
$$= s_y^2-2b\,s_{xy}+b^2s_x^2 = s_y^2-b^2s_x^2,$$

since $s_{xy}=bs_x^2$. Moreover $\hat{\rho}=\frac{s_{xy}}{s_xs_y}=\frac{bs_x^2}{s_xs_y}$ gives $\hat{\rho}s_y=bs_x$, so $s_y^2-b^2s_x^2=s_y^2(1-\hat{\rho}^2)$.

3.1 Mean versus ratio

From Ch. 5,

$$\widehat{\mathrm{RE}}(\hat{\mu}_y/\bar{y})=\frac{s_y^2}{s_y^2+r^2s_x^2-2r\hat{\rho}s_xs_y},$$

and ratio beats $\bar{y}$ iff $\hat{\rho}>\frac{1}{2}\cdot\frac{s_x/\bar{x}}{s_y/\bar{y}}$; r's bias is negligible for a line through the origin.

3.2 Mean versus regression

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\bar{y})=\frac{s_y^2}{s_y^2(1-\hat{\rho}^2)}=\frac{1}{1-\hat{\rho}^2}>1 \quad \text{unless } \hat{\rho}=0.$$

The regression estimator is (asymptotically) never worse than $\bar{y}$.

3.3 Ratio versus regression

Theorem 7.6

$$\widehat{\mathrm{RE}}(\hat{\mu}_{yL}/\hat{\mu}_y)=\frac{s_y^2+r^2s_x^2-2r\hat{\rho}s_xs_y}{s_y^2(1-\hat{\rho}^2)}\ \geq 1,$$

with equality iff $r=b$.

Proof. The numerator minus the denominator is

$$r^2s_x^2-2r\hat{\rho}s_xs_y+\hat{\rho}^2s_y^2=(rs_x-\hat{\rho}s_y)^2=(rs_x-bs_x)^2=s_x^2(r-b)^2\geq 0.$$

■

Both are linear in y: $r=\sum d_iy_i$, $d_i=1/\sum x_j$, and $b=\sum w_iy_i$, $w_i=(x_i-\bar{x})/\sum(x_j-\bar{x})^2$. They coincide only when the LS line passes through the origin, exactly when ratio is as good as regression.

4 Selected exercises

Motivation. Regression estimation in stratified sampling (separate and combined), and the exact design bias of $\hat{\mu}_{yL}$ as a reweighting.

Exercise 7.1 (Separate regression estimator)

In stratified random sampling with strata $h = 1, \dots, L$, weights w_h and known stratum means μ_{xh}, the separate regression estimator is

$$\hat{\mu}_{yLS} = \sum_{h=1}^{L} w_h [\bar{y}_h + b_h(\mu_{xh} - \bar{x}_h)],$$

with b_h the LS slope in stratum h. Find its approximate variance and an estimator of it.

Solution.

Step 1. (Independence.) The bracket $\hat{\mu}_{yL,h}$ is the regression estimator of stratum h; strata are independent, so

$$\mathrm{Var}(\hat{\mu}_{yLS}) = \sum_h w_h^2 \mathrm{Var}(\hat{\mu}_{yL,h}).$$

Step 2. (Each stratum.) By the estimated-slope argument of this chapter applied in stratum h, $b_h \approx \beta_h = S_{xyh}/S_{xh}^2$ for large n_h, and the fixed-slope theorem gives

$$\mathrm{Var}(\hat{\mu}_{yL,h}) \approx \frac{1 - f_h}{n_h} S_{yh}^2 (1 - \rho_h^2), \qquad \rho_h = \frac{S_{xyh}}{S_{xh} S_{yh}}.$$

Step 3. Hence

$$\mathrm{Var}(\hat{\mu}_{yLS}) \approx \sum_{h=1}^{L} w_h^2 \frac{1 - f_h}{n_h} S_{yh}^2 (1 - \rho_h^2), \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{yLS}) = \sum_{h=1}^{L} w_h^2 \frac{1 - f_h}{n_h} MS_{E,h},$$

where $MS_{E,h}$ is the residual mean square of the regression of y on x in stratum h.

Step 4. As for the separate ratio estimator, Step 2 needs large n_h; with small strata the slope errors and biases accumulate. ■

Exercise 7.2 (Combined regression estimator with a given slope)

With a slope β fixed in advance (the same for all strata), let $\hat{\mu}_{yLC} = \bar{y}_{\mathrm{st}} + \beta(\mu_x - \bar{x}_{\mathrm{st}})$.
(a) Show that $\hat{\mu}_{yLC}$ is unbiased and find $\mathrm{Var}(\hat{\mu}_{yLC})$.
(b) Find the β minimizing the variance and compare $\hat{\mu}_{yLC}$ with the combined ratio estimator $\hat{\mu}_{yRC}$.

Solution. Write $a_h = w_h^2(1 - f_h)/n_h$ and S_{xyh} for the covariance of x and y in stratum h.

(a)

Step 1. $\hat{\mu}_{yLC} = \sum_h w_h \bar{u}_h + \beta\mu_x$ is the stratified mean of $u_{hi} = y_{hi} - \beta x_{hi}$ plus a constant. Since $\mathrm{E}(\bar{y}_{\mathrm{st}}) = \mu_y$ and $\mathrm{E}(\bar{x}_{\mathrm{st}}) = \mu_x$, $\mathrm{E}(\hat{\mu}_{yLC}) = \mu_y$ exactly.

Step 2. By the stratified variance theorem applied to u, whose stratum variance is $S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2$,

$$\mathrm{Var}(\hat{\mu}_{yLC}) = Q(\beta) := \sum_h a_h (S_{yh}^2 - 2\beta S_{xyh} + \beta^2 S_{xh}^2) = A - 2\beta B + \beta^2 C,$$

with $A = \sum_h a_h S_{yh}^2$, $B = \sum_h a_h S_{xyh}$, $C = \sum_h a_h S_{xh}^2 > 0$.

(b)

Step 1. Q is minimized at $\beta_{\mathrm{opt}} = B/C = \sum_h a_h S_{xyh} / \sum_h a_h S_{xh}^2$, a weighted average of the within-stratum covariances over variances, and completing the square gives

$$Q(\beta) = Q(\beta_{\mathrm{opt}}) + C(\beta - \beta_{\mathrm{opt}})^2.$$

Step 2. By Ch. 5, $\mathrm{Var}(\hat{\mu}_{yRC}) \approx \sum_h a_h S_{h,C}^2$, $S_{h,C}^2 = S_{yh}^2 - 2R_C\rho_h S_{xh}S_{yh} + R_C^2 S_{xh}^2$, i.e. $Q(R_C)$: to first order, combined regression with slope $R_C = \mu_y/\mu_x$.

Step 3. Therefore

$$\mathrm{Var}(\hat{\mu}_{yRC}) - \mathrm{Var}(\hat{\mu}_{yLC})\big|_{\beta=\beta_{\mathrm{opt}}} \approx C\,(R_C - \beta_{\mathrm{opt}})^2 \;\geq 0,$$

with equality iff $R_C = \beta_{\mathrm{opt}}$ (pooled within-stratum line through the origin). Estimating β_{opt} by the pooled slope changes the variance only at smaller order. ■

Exercise 7.3 (Exact bias of the regression estimator)

Under SRS, assume the sample x's are never all equal, so b is defined. Show that

$$\mathrm{E}(\hat{\mu}_{yL}) - \mu_y = -\mathrm{Cov}(b, \bar{x}), \qquad |\mathrm{E}(\hat{\mu}_{yL}) - \mu_y| \leq \sqrt{\mathrm{Var}(b)}\,\sqrt{\frac{1-f}{n}}\,S_x,$$

and explain why the bias is negligible relative to the standard error for large n.

Solution. [Supplement]

Step 1. $\mathrm{E}(\bar{y}) = \mu_y$ and $\mathrm{E}(\bar{x} - \mu_x) = 0$ under SRS, so

$$\mathrm{E}(\hat{\mu}_{yL}) - \mu_y = -\mathrm{E}\big[b(\bar{x} - \mu_x)\big] = -\big\{\mathrm{E}\big[b(\bar{x} - \mu_x)\big] - \mathrm{E}(b)\,\mathrm{E}(\bar{x} - \mu_x)\big\} = -\mathrm{Cov}(b, \bar{x}).$$

Step 2. By the Cauchy–Schwarz inequality $|\mathrm{Cov}(b, \bar{x})| \leq \sqrt{\mathrm{Var}(b)\mathrm{Var}(\bar{x})}$, and $\mathrm{Var}(\bar{x}) = \frac{1-f}{n}S_x^2$.

Step 3. $\sqrt{\mathrm{Var}(b)}$, $\sqrt{\mathrm{Var}(\bar{x})}$ are $O(n^{-1/2})$, so the bias is $O(1/n)$ against an $O(n^{-1/2})$ SE, as for the ratio bound $|\mathrm{E}(r) - R| \leq CV(\bar{x})\sqrt{\mathrm{Var}(r)}$. It vanishes when b is uncorrelated with $\bar{x}$, e.g. under the linear model with fixed x. ■

Exercise 7.4 (The regression estimator as a calibrated weighting)

Let $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 > 0$ and $g_i = 1 + \dfrac{n(\mu_x - \bar{x})(x_i - \bar{x})}{S_{xx}}$.

(a) Show that $\hat{\mu}_{yL} = \frac{1}{n}\sum_i g_i y_i$.

(b) Show that $\frac{1}{n}\sum_i g_i = 1$ and $\frac{1}{n}\sum_i g_i x_i = \mu_x$: the weights reproduce the known mean of x exactly.

(c) Show that, among all weights (g_i') satisfying the two equations in (b), the g_i minimize $\sum_i (g_i' - 1)^2$.

Solution. [Supplement]

(a) $b = \sum_i (x_i - \bar{x})y_i / S_{xx}$, so $\bar{y} + b(\mu_x - \bar{x}) = \frac{1}{n}\sum_i y_i\big[1 + n(\mu_x - \bar{x})(x_i - \bar{x})/S_{xx}\big] = \frac{1}{n}\sum_i g_i y_i$.

(b) Since $\sum_i (x_i - \bar{x}) = 0$, $\sum_i g_i = n$. Since $\sum_i (x_i - \bar{x})x_i = S_{xx}$, $\sum_i g_i x_i = n\bar{x} + n(\mu_x - \bar{x}) = n\mu_x$.

(c)

Step 1. Put $h_i = g_i - 1 = \lambda(x_i - \bar{x})$ with $\lambda = n(\mu_x - \bar{x})/S_{xx}$, and let g_i' be any feasible weights, $d_i = g_i' - g_i$. Subtracting the constraints for g' and g, $\sum_i d_i = 0$ and $\sum_i d_i x_i = 0$.

Step 2. Then $\sum_i h_i d_i = \lambda\big(\sum_i d_i x_i - \bar{x}\sum_i d_i\big) = 0$, so

$$\sum_i (g_i' - 1)^2 = \sum_i (h_i + d_i)^2 = \sum_i h_i^2 + \sum_i d_i^2 \geq \sum_i h_i^2,$$

with equality iff $d = 0$.

So $\hat{\mu}_{yL}$ is the weighted mean with weights closest to 1 calibrated to μ_x; if $\bar{x} < \mu_x$, large-x units get weights above 1 (the upward correction). ■

Chapter 8. Cluster Sampling

1 Introduction

For weekly food expenditure in Seoul, element sampling needs a list of all families (rarely available) and scatters interviews over the city. A list of blocks is easy to get, and interviewing every family in a few blocks is cheap.

Definition 8.1

A **cluster sample** is an SRS of clusters of elements, observing every element of a selected cluster.

Cluster sampling is effective when

- a good frame of elements is unavailable or costly, while a frame of clusters is easy to obtain;
- the cost of observation increases with the distance between elements.

Since it is SRS of clusters, the estimators parallel those of Ch. 2, with cluster totals playing the role of observations.

Notation.

N	number of clusters in the population
n	number of clusters selected by SRS
m_i	number of elements in cluster i, $i = 1, \dots, N$
$\bar{m} = \frac{1}{n}\sum_{i=1}^{n} m_i$	average cluster size in the sample
$M = \sum_{i=1}^{N} m_i$	number of elements in the population
$\bar{M} = M/N$	average cluster size in the population
y_{ij}	value of the jth element in cluster i
$y_i = \sum_{j=1}^{m_i} y_{ij}$	total of cluster i

2 Estimation of the Population Mean

The population mean per element is

$$\mu = \frac{\sum_{i=1}^{N}\sum_{j=1}^{m_i} y_{ij}}{\sum_{i=1}^{N} m_i} = \frac{\sum_{i=1}^{N} y_i}{M},$$

a ratio of the cluster totals y_i to the cluster sizes m_i. Its natural estimator is therefore a ratio estimator (Ch. 5) with $x_i = m_i$.

Theorem 8.2: Cluster estimator of μ

$$\bar{y}_{\text{cl}} = \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}\, s_c^2, \qquad s_c^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y}_{\text{cl}} m_i)^2,$$

where $\bar{M}$ is replaced by $\bar{m}$ if M is unknown. An approximate 95% interval is $\bar{y}_{\text{cl}} \pm 2\sqrt{\widehat{\text{Var}}(\bar{y}_{\text{cl}})}$.

Derivation: from the ratio estimator

With $x_i = m_i$, $R = \mu$ and $\mu_x = \bar{M}$, Ch. 5 gives

$$\mathrm{Var}(\bar{y}_{\mathrm{cl}}) \approx \frac{1}{\bar{M}^2} \cdot \frac{N-n}{N} \cdot \frac{1}{n} \cdot \frac{1}{N-1} \sum_{i=1}^{N} (y_i - \mu m_i)^2 = \frac{1}{\bar{M}^2} \frac{N-n}{Nn} (S_y^2 + \mu^2 S_m^2 - 2\mu\rho S_m S_y),$$

and replacing population quantities by sample ones yields $\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}})$ and the alternative form

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{N-n}{Nn\bar{M}^2} (s_y^2 + \bar{y}_{\mathrm{cl}}^2 s_m^2 - 2\bar{y}_{\mathrm{cl}} \hat{\rho} s_m s_y), \qquad s_m^2 = \frac{1}{n-1} \sum_{i=1}^{n} (m_i - \bar{m})^2,$$

where s_y^2 is the sample variance of the cluster totals. Cluster totals nearly proportional to cluster sizes ($y_i \approx \mu m_i$) give a small variance.

3 Estimation of the Population Total

3.1 M known

Since $\tau = M\mu$,

$$\hat{\tau} = M\bar{y}_{\mathrm{cl}} = M \frac{\sum_{i=1}^{n} y_i}{\sum_{i=1}^{n} m_i}, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = M^2 \widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = N^2 \frac{N-n}{Nn} s_c^2.$$

3.2 M unknown

Often the number of elements is unknown. Then use the mean of the cluster totals,

$$\bar{y}_t = \frac{1}{n} \sum_{i=1}^{n} y_i, \qquad \mathrm{E}(\bar{y}_t) = \frac{1}{N} \sum_{i=1}^{N} y_i = \frac{\tau}{N},$$

which is unbiased for the average cluster total, so $N\bar{y}_t$ is unbiased for τ:

$$\hat{\tau} = N\bar{y}_t = \frac{N}{n} \sum_{i=1}^{n} y_i, \qquad \widehat{\mathrm{Var}}(\hat{\tau}) = N^2 \frac{N-n}{Nn} s_t^2, \qquad s_t^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y}_t)^2.$$

This SRS of cluster totals ignores m_i; it is worse than $M\bar{y}_{\mathrm{cl}}$ when the totals vary mainly through the sizes.

If all $m_i = m$, then $M = Nm$, $\bar{y}_{\mathrm{cl}} = \bar{y}_t/m$, and the two estimators of τ coincide.

4 Comparison with Simple Random Sampling

Assume equal cluster sizes $m_i = m$, so $M = Nm$ and

$$\bar{y}_{\mathrm{cl}} = \frac{\sum_{i=1}^{n} y_i}{nm} = \frac{1}{nm} \sum_{i=1}^{n} \sum_{j=1}^{m} y_{ij} = \frac{\bar{y}_t}{m}.$$

Let $\bar{y}_i = y_i/m$ be the mean of cluster i. Then

$$\widehat{\mathrm{Var}}(\bar{y}_{\mathrm{cl}}) = \frac{1}{m^2} \widehat{\mathrm{Var}}(\bar{y}_t) = \frac{1}{m^2} \frac{N-n}{Nn} \frac{1}{n-1} \sum_{i} (m\bar{y}_i - m\bar{y}_{\mathrm{cl}})^2 = \frac{N-n}{Nn} \frac{1}{n-1} \sum_{i=1}^{n} (\bar{y}_i - \bar{y}_{\mathrm{cl}})^2.$$

Treat the nm observed elements as an $m \times n$ table with column means $\bar{y}_i$ and grand mean $\bar{y}_{\text{cl}}$:

$$\underbrace{\sum_{i=1}^{n}\sum_{j=1}^{m}(y_{ij}-\bar{y}_{\text{cl}})^2}_{SST} = \underbrace{\sum_{i}\sum_{j}(y_{ij}-\bar{y}_i)^2}_{SSW} + \underbrace{m\sum_{i}(\bar{y}_i-\bar{y}_{\text{cl}})^2}_{SSB},$$

with $MSB = SSB/(n-1)$ and $MSW = SSW/[n(m-1)]$. Then

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{N-n}{Nnm}\,MSB.$$

If the same nm elements had been drawn by SRS from the Nm elements, with sample variance s^2,

$$\widehat{\text{Var}}(\bar{y}) = \frac{Nm-nm}{Nm}\cdot\frac{s^2}{nm} = \frac{N-n}{Nnm}\,s^2.$$

Remark 8.3: Efficiency of cluster sampling

Compare MSB with s^2: cluster sampling is more efficient when clusters are miniatures of the population (heterogeneous within). Usually neighbours are alike, $MSB > s^2$, and it is chosen for lower cost per element.

5 Selecting the Sample Size

Setting $B = 2\sqrt{\frac{N-n}{Nn\bar{M}^2}s_c^2}$ and solving as in Ch. 2:

- for μ with $\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND+s_c^2}$, $D = \dfrac{B^2\bar{M}^2}{4}$;
- for τ with $M\bar{y}_{\text{cl}}$: $n = \dfrac{Ns_c^2}{ND+s_c^2}$, $D = \dfrac{B^2}{4N^2}$;
- for τ with $N\bar{y}_t$: $n = \dfrac{Ns_t^2}{ND+s_t^2}$, $D = \dfrac{B^2}{4N^2}$,

with s_c^2, s_t^2 taken from a previous survey or a pilot sample.

6 Estimation of a Population Proportion

Let a_i be the number of elements in cluster i having the characteristic. Then p is the ratio $\sum a_i/\sum m_i$ over the population, and

$$\hat{p}_{\text{cl}} = \frac{\sum_{i=1}^{n}a_i}{\sum_{i=1}^{n}m_i}, \qquad \widehat{\text{Var}}(\hat{p}_{\text{cl}}) = \frac{N-n}{Nn\bar{M}^2}s_p^2, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^{n}(a_i-\hat{p}_{\text{cl}}m_i)^2,$$

with $\bar{M} \approx \bar{m}$ if M is unknown. The sample size for a bound B is $n = Ns_p^2/(ND+s_p^2)$ with $D = B^2\bar{M}^2/4$.

7 Cluster Sampling Combined with Stratification

Books read by 4th and 5th graders: strata are grades, clusters are classes. In stratum ℓ, N_ℓ clusters, $\bar{y}_{t\ell}$ the mean sampled cluster total, $\bar{m}_\ell$ the mean sampled size:

$$\text{average cluster total: } \frac{1}{N}(N_1\bar{y}_{t1} + N_2\bar{y}_{t2}), \qquad \text{average cluster size: } \frac{1}{N}(N_1\bar{m}_1 + N_2\bar{m}_2),$$

$$\bar{y}_c = \frac{N_1\bar{y}_{t1} + N_2\bar{y}_{t2}}{N_1\bar{m}_1 + N_2\bar{m}_2} = \frac{\sum_\ell N_\ell \bar{y}_{t\ell}}{\sum_\ell N_\ell \bar{m}_\ell},$$

a combined ratio estimator. Its estimated variance is

$$\widehat{\mathrm{Var}}(\bar{y}_c) = \frac{1}{\left(\sum_\ell N_\ell \bar{m}_\ell\right)^2} \sum_{\ell=1}^{L} N_\ell(N_\ell - n_\ell)\frac{s_{t\ell}^2}{n_\ell},$$

with $s_{t\ell}^2$ the sample variance of cluster totals in stratum ℓ: the numerator is $\sum_\ell N_\ell^2 \widehat{\mathrm{Var}}(\bar{y}_{t\ell})$, the denominator the squared estimated number of elements.

8 Cluster Sampling with Probabilities Proportional to Size

8.1 The PPS estimator

Sampling with replacement from $y_1, \ldots, y_N$, unit i drawn with probability z_i each draw ($\sum z_i = 1$).

Theorem 8.4: Hansen–Hurwitz estimator

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{z_i}$$

is unbiased for τ for any choice of $z_i > 0$, with $\widehat{\mathrm{Var}}(\hat{\tau}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}\left(\frac{y_i}{z_i} - \hat{\tau}_{\text{pps}}\right)^2$.

Proof. Each draw gives an iid $t = y_I/z_I$, $P(I = k) = z_k$, with $\mathrm{E}(t) = \tau$; the mean of n draws is unbiased with variance estimated by s_t^2/n. ■

With $z_i = 1/N$ (equal probability), $\hat{\tau}_{\text{pps}} = \frac{N}{n}\sum y_i = N\bar{y}$, the usual estimator. The variance is small when $z_i \propto y_i$, so we want z_i roughly proportional to y_i.

8.2 Application to clusters

Cluster totals tend to be proportional to cluster sizes, so choose $z_i = m_i/M$. With y_i the cluster total and $\bar{y}_i = y_i/m_i$ the cluster mean,

$$\hat{\tau}_{\text{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{y_i}{m_i/M} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\text{pps}} = \frac{\hat{\tau}_{\text{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i :$$

the estimator of μ is the plain average of the sampled cluster means.

Theorem 8.5

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\text{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\text{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\text{pps}})^2.$$

Proof. The $\bar{y}_i$ are iid draws (with replacement), so $\widehat{\mathrm{Var}}(\frac{1}{n}\sum\bar{y}_i) = \frac{1}{n}\cdot\frac{1}{n-1}\sum(\bar{y}_i - \hat{\mu}_{\text{pps}})^2$; multiply by M^2 for the total. ■

Remark 8.6: Equal sizes

If $m_i = m$, then $\hat{\mu}_{\text{pps}} = \frac{1}{n}\sum_i \frac{1}{m}\sum_j y_{ij} = \bar{y}_{\text{cl}} = \bar{y}_t/m$, and since $s_t^2 = \frac{m^2}{n-1}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2$,

$$\widehat{\text{Var}}(\bar{y}_{\text{cl}}) = \frac{1}{m^2}\frac{N-n}{Nn}s_t^2 = \frac{N-n}{N}\cdot\frac{1}{n(n-1)}\sum(\bar{y}_i - \bar{y}_{\text{cl}})^2,$$

which is $\widehat{\text{Var}}(\hat{\mu}_{\text{pps}})$ up to the fpc (PPS draws are with replacement).

9 Selected exercises

Motivation. The exact design effect $1 + (m-1)\rho$, and the exact PPS variance showing that size-proportional selection removes size variation from the error.

Exercise 8.1 (Design effect of cluster sampling)

Let all N clusters have m elements ($M = Nm$), let μ and $S^2 = \frac{1}{Nm-1}\sum_{i=1}^{N}\sum_{j=1}^{m}(y_{ij}-\mu)^2$ be the element mean and variance, and define the intracluster correlation

$$\rho = \frac{\sum_{i=1}^{N}\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)}{(m-1)(Nm-1)S^2}.$$

For an SRS of n clusters, $f = n/N$, show that

$$\text{Var}(\bar{y}_{\text{cl}}) = \frac{1-f}{nm}S^2\cdot\frac{Nm-1}{m(N-1)}[1+(m-1)\rho],$$

and that the ratio of $\text{Var}(\bar{y}_{\text{cl}})$ to the variance of the mean of an SRS of nm elements is approximately $1+(m-1)\rho$ when N is large.

Solution. [Supplement]

Step 1. With equal sizes, $\bar{y}_{\text{cl}} = \frac{1}{n}\sum_{i\le n}\bar{y}_i$ is the SRS mean of the N cluster means $\bar{y}_i = \frac{1}{m}\sum_j y_{ij}$, whose population mean is μ. By Ch. 2,

$$\text{Var}(\bar{y}_{\text{cl}}) = \frac{1-f}{n}S_b^2, \qquad S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\bar{y}_i-\mu)^2.$$

Step 2. Expanding a square of a sum,

$$\sum_{i=1}^{N}(\bar{y}_i-\mu)^2 = \frac{1}{m^2}\sum_i\Big(\sum_j(y_{ij}-\mu)\Big)^2 = \frac{1}{m^2}\Big[\sum_i\sum_j(y_{ij}-\mu)^2 + \sum_i\sum_{j\neq j'}(y_{ij}-\mu)(y_{ij'}-\mu)\Big]$$
$$= \frac{1}{m^2}\Big[(Nm-1)S^2 + (m-1)(Nm-1)S^2\rho\Big] = \frac{(Nm-1)S^2}{m^2}[1+(m-1)\rho].$$

Step 3. Substituting into Step 1 gives the stated formula.

Step 4. (Design effect.) An SRS of nm of the Nm elements has sampling fraction $nm/(Nm) = f$ and variance $\frac{1-f}{nm}S^2$. The ratio is $\frac{Nm-1}{m(N-1)}[1+(m-1)\rho]$, and $\frac{Nm-1}{m(N-1)} \to 1$ as $N\to\infty$.

With $\rho > 0$, nm cluster elements are worth about $nm/[1+(m-1)\rho]$ independent ones; $\rho = 0.05$ doubles the variance for $m = 21$. Efficiency per element needs $\rho \le 0$. ■

Exercise 8.2 (Exact variance of the PPS estimator)

In the setting of the Hansen–Hurwitz theorem (draws with replacement, selection probabilities $z_k > 0$, $\sum z_k = 1$), show that
(a) $\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \dfrac{1}{n}\sum_{k=1}^{N} z_k\Big(\dfrac{y_k}{z_k} - \tau\Big)^2$;
(b) if all $y_k > 0$, the variance is zero iff $z_k = y_k/\tau$ for all k;
(c) $\widehat{\mathrm{Var}}(\hat\tau_{\mathrm{pps}})$ of the theorem is unbiased for $\mathrm{Var}(\hat\tau_{\mathrm{pps}})$;
(d) for clusters with $z_i = m_i/M$, $\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \dfrac{M}{n}\sum_{i=1}^{N} m_i(\bar y_i - \mu)^2$, where $\bar y_i$ is the mean of cluster i.

Solution. [Supplement] Let $t_r = y_{I_r}/z_{I_r}$, $r = 1, \dots, n$, where $I_1, \dots, I_n$ are the independent draws, $P(I_r = k) = z_k$. Then $\hat\tau_{\mathrm{pps}} = \bar t$.

(a) The t_r are iid with mean τ (proof of the theorem) and variance $V_1 = \mathrm{E}(t_1 - \tau)^2 = \sum_k z_k(y_k/z_k - \tau)^2$. Hence $\mathrm{Var}(\bar t) = V_1/n$.

(b) $V_1 = 0$ iff $y_k/z_k = \tau$ for all k, i.e. $z_k = y_k/\tau$; infeasible (needs the y_k), but variance is small for z_k nearly $\propto y_k$.

(c) For iid t_r the sample variance $s_t^2 = \frac{1}{n-1}\sum_r (t_r - \bar t)^2$ is unbiased for V_1 (the with-replacement case of Ch. 2). Hence $\mathrm{E}\big[s_t^2/n\big] = V_1/n = \mathrm{Var}(\hat\tau_{\mathrm{pps}})$, and s_t^2/n is the $\widehat{\mathrm{Var}}(\hat\tau_{\mathrm{pps}})$ of the theorem.

(d) With y_i the cluster total, $y_i/z_i = My_i/m_i = M\bar y_i$ and $\tau = M\mu$. By (a),

$$\mathrm{Var}(\hat\tau_{\mathrm{pps}}) = \frac{1}{n}\sum_i \frac{m_i}{M} M^2(\bar y_i - \mu)^2 = \frac{M}{n}\sum_i m_i(\bar y_i - \mu)^2.$$

Only cluster *means* vary here; $N\bar y_t$ is driven by cluster *totals* $y_i = m_i\bar y_i$, which vary also through the sizes. ■

Chapter 9. Two-Stage Cluster Sampling

1 Introduction

Definition 9.1

A **two-stage cluster sample** draws a probability sample of clusters (**primary sampling units**, psu, e.g. classes), then of elements within them (**secondary sampling units**, ssu, e.g. students).

When elements in a cluster are similar, subsampling spreads the budget over more clusters.

Notation. Both stages use SRS without replacement.

N, n	number of clusters in the population / selected
M_i, m_i	number of elements in cluster i / selected from it
$M = \sum_{i=1}^{N} M_i$, $\bar{M} = M/N$	population size and average cluster size
y_{ij}	value of the jth sampled element of cluster i
$\bar{y}_i = \frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}$	sample mean of cluster i
μ_i, $\tau_i = M_i\mu_i$	true mean and total of cluster i; $\tau = \sum_{i=1}^{N}\tau_i$
$\hat{\tau}_i = M_i\bar{y}_i$	estimated total of cluster i
$S_i^2 = \frac{1}{M_i-1}\sum_{j=1}^{M_i}(y_{ij}-\mu_i)^2$	within-cluster variance
$S_b^2 = \frac{1}{N-1}\sum_{i=1}^{N}(\tau_i-\bar{\tau})^2$, $\bar{\tau} = \tau/N$	between-cluster variance of totals

2 Estimation of the Total and the Mean

Motivation. Subsampling large clusters is cheaper; the variance has between- and within-cluster components.

Theorem 9.2: Unbiased estimators

$$\hat{\tau} = \frac{N}{n}\sum_{i=1}^{n} M_i\bar{y}_i = \frac{N}{n}\sum_{i=1}^{n} M_i\frac{1}{m_i}\sum_{j=1}^{m_i} y_{ij}, \qquad \bar{y} = \frac{\hat{\tau}}{M}$$

are unbiased for τ and μ respectively.

Proof. With E_2 over the second stage (clusters fixed) and E_1 over the first: $\mathrm{E}_2(\bar{y}_i) = \mu_i$, so $\mathrm{E}_2(\hat{\tau}_i) = \tau_i$ and

$$\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n}\tau_i.$$

This is N times an SRS mean of the cluster totals, so $\mathrm{E}_1\mathrm{E}_2(\hat{\tau}) = N\cdot\frac{\tau}{N} = \tau$. Then $\mathrm{E}(\bar{y}) = \tau/M = \mu$. ■

Theorem 9.3: Variance

With $u_i = M_i^2\,\frac{M_i-m_i}{M_i}\,\frac{S_i^2}{m_i}$ (the second-stage variance of $\hat{\tau}_i$),

$$\mathrm{Var}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{S_b^2}{n} + \frac{N}{n}\sum_{i=1}^{N} u_i = \underbrace{N^2\frac{N-n}{N}\frac{S_b^2}{n}}_{\text{between psu}} + \underbrace{\frac{N}{n}\sum_{i=1}^{N} M_i^2\frac{M_i-m_i}{M_i}\frac{S_i^2}{m_i}}_{\text{within psu}}.$$

2.1 Variance of a two-stage estimator

Theorem 9.4: Conditional variance decomposition

For any estimator $\hat{\theta}$ computed from a two-stage sample,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1\big(\mathrm{E}_2(\hat{\theta})\big) + \mathrm{E}_1\big(\mathrm{Var}_2(\hat{\theta})\big).$$

For three stages, $\mathrm{Var}(\hat{\theta}) = \mathrm{Var}_1(\mathrm{E}_2\mathrm{E}_3\hat{\theta}) + \mathrm{E}_1\big(\mathrm{Var}_2(\mathrm{E}_3\hat{\theta})\big) + \mathrm{E}_1\mathrm{E}_2\big(\mathrm{Var}_3\hat{\theta}\big)$.

Proof. Let $\theta = \mathrm{E}(\hat{\theta})$. Conditionally on the first stage,

$$\mathrm{E}_2(\hat{\theta} - \theta)^2 = \mathrm{Var}_2(\hat{\theta}) + \big(\mathrm{E}_2\hat{\theta}\big)^2 - 2\theta\,\mathrm{E}_2\hat{\theta} + \theta^2.$$

Taking E_1 and using $\mathrm{E}_1\mathrm{E}_2\hat{\theta} = \theta$,

$$\mathrm{Var}(\hat{\theta}) = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{E}_1\big(\mathrm{E}_2\hat{\theta}\big)^2 - \theta^2 = \mathrm{E}_1\mathrm{Var}_2(\hat{\theta}) + \mathrm{Var}_1\big(\mathrm{E}_2\hat{\theta}\big).$$

■

Derivation: proof of the variance formula

Between term. $\mathrm{E}_2(\hat{\tau}) = \frac{N}{n}\sum_{i=1}^{n}\tau_i$ is N times the SRS mean of the N cluster totals, so by Ch. 2

$$\mathrm{Var}_1\big(\mathrm{E}_2\hat{\tau}\big) = N^2\frac{N-n}{N}\frac{S_b^2}{n}.$$

Within term. Given the clusters, the subsamples are independent, and $\mathrm{Var}_2(\hat{\tau}_i) = M_i^2\mathrm{Var}_2(\bar{y}_i) = M_i^2\frac{M_i-m_i}{M_i}\frac{S_i^2}{m_i} = u_i$. Thus

$$\mathrm{Var}_2(\hat{\tau}) = \frac{N^2}{n^2}\sum_{i=1}^{n}u_i, \qquad \mathrm{E}_1\mathrm{Var}_2(\hat{\tau}) = \frac{N^2}{n^2}\cdot n\cdot\frac{1}{N}\sum_{i=1}^{N}u_i = \frac{N}{n}\sum_{i=1}^{N}u_i,$$

because each cluster is in the first-stage sample with probability n/N.

Direct expansion. Square $\hat{\tau} - \tau = \big(\frac{N}{n}\sum_{i\le n}\tau_i - \tau\big) + \frac{N}{n}\sum_{i\le n}(\hat{\tau}_i - \tau_i)$: the cross term vanishes ($\mathrm{E}_2(\hat{\tau}_i - \tau_i) = 0$), as do products for $i \neq i'$ (independent subsamples); the squares give the two components.

2.2 Estimated variance

Theorem 9.5: Unbiased variance estimator

$$\widehat{\mathrm{Var}}(\hat{\tau}) = N^2\frac{N-n}{N}\frac{s_b^2}{n} + \frac{N}{n}\sum_{i=1}^{n}M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i},$$

$$s_b^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar{y}_i - \bar{M}\bar{y})^2, \qquad s_i^2 = \frac{1}{m_i-1}\sum_{j=1}^{m_i}(y_{ij} - \bar{y}_i)^2,$$

and $\widehat{\mathrm{Var}}(\bar{y}) = \widehat{\mathrm{Var}}(\hat{\tau})/M^2$. Here s_b^2 is the sample variance of the $\hat{\tau}_i = M_i\bar{y}_i$ (note $\bar{M}\bar{y} = \hat{\tau}/N = \frac{1}{n}\sum\hat{\tau}_i$).

The key fact is that s_b^2 is *not* unbiased for S_b^2: the estimated cluster totals also carry second-stage noise.

Theorem 9.6

$\mathrm{E}(s_b^2) = S_b^2 + \dfrac{1}{N}\sum_{i=1}^{N} u_i$, and consequently $\mathrm{E}[\widehat{\mathrm{Var}}(\hat\tau)] = \mathrm{Var}(\hat\tau)$.

Derivation: proof

Write $\bar\tau = \frac{1}{n}\sum_{i\le n}\hat\tau_i = \hat\tau/N$. Then $(n-1)s_b^2 = \sum_{i\le n}\hat\tau_i^2 - n\bar\tau^2$.
First piece. $\mathrm{E}_2(\hat\tau_i^2) = \mathrm{Var}_2(\hat\tau_i) + \tau_i^2 = u_i + \tau_i^2$, so $\mathrm{E}\big[\sum_{i\le n}\hat\tau_i^2\big] = \frac{n}{N}\sum_{i=1}^{N}(u_i+\tau_i^2)$.
Second piece. $\mathrm{E}(\bar\tau^2) = \frac{\mathrm{Var}(\hat\tau)}{N^2} + \bar\tau^2 = \frac{N-n}{Nn}S_b^2 + \frac{1}{nN}\sum_{i=1}^{N}u_i + \bar\tau^2$.
Combining and using $\frac{1}{N}\sum_{i=1}^{N}\tau_i^2 - \bar\tau^2 = \frac{N-1}{N}S_b^2$,

$$\begin{aligned}\mathrm{E}\big[(n-1)s_b^2\big] &= \frac{n}{N}\sum u_i + n\Big(\frac{1}{N}\sum\tau_i^2 - \bar\tau^2\Big) - \frac{N-n}{N}S_b^2 - \frac{1}{N}\sum u_i \\ &= \frac{n-1}{N}\sum_{i=1}^{N}u_i + \frac{n(N-1)-(N-n)}{N}S_b^2 = (n-1)\Big[S_b^2 + \frac{1}{N}\sum_{i=1}^{N}u_i\Big].\end{aligned}$$

Unbiasedness of $\widehat{\mathrm{Var}}(\hat\tau)$. Since $\mathrm{E}_2(s_i^2) = S_i^2$, the second term has expectation $\frac{N}{n}\cdot\frac{n}{N}\sum_{i=1}^{N}u_i = \sum_{i=1}^{N}u_i$. Hence

$$\mathrm{E}\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{Nn}\Big(S_b^2 + \frac{1}{N}\sum u_i\Big) + \sum u_i = N^2\frac{N-n}{Nn}S_b^2 + \Big(\frac{N-n}{n}+1\Big)\sum u_i = \mathrm{Var}(\hat\tau).$$

Of the within-psu variance $\frac{N}{n}\sum u_i$, the between-psu term of $\widehat{\mathrm{Var}}$ already contributes $\frac{N-n}{n}\sum u_i$, and the explicit within term supplies the remaining $\sum u_i$.

Remark 9.7: Special cases

- All elements of each selected psu are measured ($m_i = M_i$): $\widehat{\mathrm{Var}}(\hat\tau) = N^2\frac{N-n}{Nn}s_b^2$, which is one-stage cluster sampling (Ch. 8).
- All psu are included ($n = N$): $\widehat{\mathrm{Var}}(\hat\tau) = \sum_{i=1}^{N} M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i}$, which is stratified random sampling with the clusters as strata (Ch. 3).

3 Estimating the Mean when M is Unknown

Motivation. With M unknown, the mean is a ratio of two estimated totals.

$\bar y = \hat\tau/M$ requires M. Otherwise use the ratio estimator

$$\hat\mu = \frac{\sum_{i=1}^{n}M_i\bar y_i}{\sum_{i=1}^{n}M_i} = \frac{\sum_{i=1}^{n}\hat\tau_i}{\sum_{i=1}^{n}M_i}.$$

Theorem 9.8

$$\widehat{\mathrm{Var}}(\hat\mu) = \frac{N^2}{M^2}\cdot\frac{N-n}{N}\cdot\frac{s_r^2}{n} + \frac{N}{nM^2}\sum_{i=1}^{n}M_i^2\frac{M_i-m_i}{M_i}\frac{s_i^2}{m_i}, \qquad s_r^2 = \frac{1}{n-1}\sum_{i=1}^{n}(M_i\bar y_i - M_i\hat\mu)^2,$$

where, when M is unknown, M is estimated by $\hat M = N\cdot\frac{1}{n}\sum_{i=1}^{n}M_i$ (equivalently $\bar M$ by the sample average of the M_i).

The only change from $\widehat{\mathrm{Var}}(\bar{y})$ is that the between term uses deviations $M_i\bar{y}_i - M_i\hat{\mu}$ (residuals from a ratio fit) instead of $M_i\bar{y}_i - \bar{M}\bar{y}$.

4 Estimation of a Proportion

For the proportion of students in favor of a law, let $y_{ij} = 1$ or 0 and $\hat{p}_i = \frac{1}{m_i}\sum_j y_{ij}$, the sample proportion in cluster i. With M unknown,

$$\hat{p} = \frac{\sum_{i=1}^n M_i\hat{p}_i}{\sum_{i=1}^n M_i},$$

$$\widehat{\mathrm{Var}}(\hat{p}) = \frac{N^2}{M^2}\frac{N-n}{N}\frac{s_p^2}{n} + \frac{N}{nM^2}\sum_{i=1}^n M_i^2\frac{M_i - m_i}{M_i}\frac{\hat{p}_i\hat{q}_i}{m_i - 1}, \qquad s_p^2 = \frac{1}{n-1}\sum_{i=1}^n (M_i\hat{p}_i - M_i\hat{p})^2,$$

with $\hat{q}_i = 1 - \hat{p}_i$. The within term uses the SRS result $\widehat{\mathrm{Var}}(\hat{p}_i) = \frac{\hat{p}_i\hat{q}_i}{m_i - 1}\frac{M_i - m_i}{M_i}$.

5 Equal-Sized Clusters

Suppose $M_1 = \cdots = M_N = \bar{M}$ and $m_1 = \cdots = m_n = m$. Then

$$\bar{y} = \frac{\hat{\tau}}{M} = \frac{1}{N\bar{M}}\cdot\frac{N}{n}\bar{M}\sum_{i=1}^n \bar{y}_i = \frac{1}{n}\sum_{i=1}^n \bar{y}_i = \frac{1}{nm}\sum_{i=1}^n\sum_{j=1}^m y_{ij} = \hat{\mu},$$

the overall sample average. Arrange the data as an $m \times n$ table (columns = psu) with

$$MSB = \frac{m}{n-1}\sum_{i=1}^n (\bar{y}_i - \bar{y})^2, \qquad MSW = \frac{1}{n(m-1)}\sum_{i=1}^n\sum_{j=1}^m (y_{ij} - \bar{y}_i)^2.$$

Theorem 9.9: Equal-size two-stage variance

With $f_1 = n/N$ and $f_2 = m/\bar{M}$,

$$\widehat{\mathrm{Var}}(\bar{y}) = (1 - f_1)\frac{MSB}{nm} + (1 - f_2)\frac{MSW}{Nm}.$$

If N is large, $\widehat{\mathrm{Var}}(\bar{y}) \approx MSB/(nm)$.

Derivation

Between term. $s_b^2 = \frac{1}{n-1}\sum(\bar{M}\bar{y}_i - \bar{M}\bar{y})^2 = \bar{M}^2\frac{MSB}{m}$, and $N^2/M^2 = 1/\bar{M}^2$, so

$$\frac{N^2}{M^2}\frac{N-n}{N}\frac{s_b^2}{n} = (1 - f_1)\frac{1}{n}\frac{MSB}{m}.$$

Within term. $\frac{1}{n}\sum_i s_i^2 = MSW$, so

$$\frac{1}{M^2}\frac{N}{n}\sum_{i=1}^n \bar{M}^2\frac{\bar{M} - m}{\bar{M}}\frac{s_i^2}{m} = \frac{N\bar{M}^2}{M^2}(1 - f_2)\frac{MSW}{m} = \frac{1}{N}(1 - f_2)\frac{MSW}{m}.$$

The same computation for $\widehat{\mathrm{Var}}(\hat{\mu})$ gives $s_r^2 = s_b^2$ and $\hat{M} = M$, so $\widehat{\mathrm{Var}}(\hat{\mu}) = \widehat{\mathrm{Var}}(\bar{y})$ here.

Remark 9.10: Limiting cases

- $m = \bar{M}$ ($f_2 = 1$): the within term vanishes and $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_1)\frac{MSB}{nm}$, one-stage cluster sampling.
- $n = N$ ($f_1 = 1$): $\widehat{\mathrm{Var}}(\bar{y}) = (1-f_2)\frac{MSW}{Nm}$, the variance estimate of a stratified random sample with N strata and m observations per stratum.

6 Probabilities Proportional to Size

PPS helps when cluster sizes vary greatly, usually at the first stage only. Selecting psu with replacement with $z_i = M_i/M$ and $\hat{\tau}_i = M_i\bar{y}_i$,

$$\hat{\tau}_{\mathrm{pps}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\hat{\tau}_i}{z_i} = \frac{M}{n}\sum_{i=1}^{n}\bar{y}_i, \qquad \hat{\mu}_{\mathrm{pps}} = \frac{\hat{\tau}_{\mathrm{pps}}}{M} = \frac{1}{n}\sum_{i=1}^{n}\bar{y}_i,$$

compared with the equal-probability choice $z_i = 1/N$, which gives $\hat{\tau} = \frac{N}{n}\sum M_i\bar{y}_i$. The variance estimators are

$$\widehat{\mathrm{Var}}(\hat{\tau}_{\mathrm{pps}}) = \frac{M^2}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2, \qquad \widehat{\mathrm{Var}}(\hat{\mu}_{\mathrm{pps}}) = \frac{1}{n(n-1)}\sum_{i=1}^{n}(\bar{y}_i - \hat{\mu}_{\mathrm{pps}})^2,$$

which include second-stage variability (each $\bar{y}_i$ is an independent draw of cluster *and* subsample). With $m_i = m$, $\hat{\mu}_{\mathrm{pps}} = \bar{y}$ and $\widehat{\mathrm{Var}} = \frac{MSB}{nm}$, the large-$N$ approximation above.

7 Selected exercises

Motivation. The PPS variance estimator captures the second stage.

Exercise 9.1 (Two-stage PPS)

Clusters are drawn n times with replacement, probabilities $z_i > 0$; each draw of cluster i takes an independent SRS of m_i of M_i elements, with $\hat{\tau}_i = M_i\bar{y}_i$ unbiased, variance $V_i = M_i^2(1 - m_i/M_i)S_i^2/m_i$. Let $t_r = \hat{\tau}_{(r)}/z_{(r)}$, $\hat{\tau}_{\mathrm{pps}} = \bar{t}$.
(a) Show that $\hat{\tau}_{\mathrm{pps}}$ is unbiased with

$$\mathrm{Var}(\hat{\tau}_{\mathrm{pps}}) = \frac{1}{n}\Big[\sum_{i=1}^{N} z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_{i=1}^{N}\frac{V_i}{z_i}\Big].$$

(b) Show that $\frac{1}{n(n-1)}\sum_r (t_r - \bar{t})^2$ is unbiased for $\mathrm{Var}(\hat{\tau}_{\mathrm{pps}})$.

Solution. [Supplement]

(a)

Step 1. (iid structure.) Each draw is an index I_r ($P(I_r = i) = z_i$) plus a fresh subsample, so the t_r are iid.

Step 2. (Mean.) Conditionally on $I_r = i$, $\mathrm{E}(t_r \mid I_r = i) = \tau_i/z_i$. Hence $\mathrm{E}t_r = \sum_i z_i\tau_i/z_i = \tau$, and $\mathrm{E}\bar{t} = \tau$.

Step 3. (Variance of one draw.) By the conditional variance decomposition, with $\mathrm{Var}(t_r \mid I_r = i) = V_i/z_i^2$,

$$\mathrm{Var}(t_r) = \mathrm{Var}(\mathrm{E}(t_r \mid I_r)) + \mathrm{E}(\mathrm{Var}(t_r \mid I_r)) = \sum_i z_i\Big(\frac{\tau_i}{z_i} - \tau\Big)^2 + \sum_i z_i\frac{V_i}{z_i^2}.$$

Dividing by n gives the formula: the first term is the one-stage PPS variance (Exercise 8.2), the second the subsampling variance.

(b) For iid t_r, $\mathrm{E}(s_t^2/n) = \mathrm{Var}(\bar{t})$: no s_i^2 is needed, yet both components are estimated, provided each repeated draw gets a new independent subsample. ■

Chapter 10. Estimating Population Size and Nonresponse

1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, mark and release n_1 fish, then catch n_2 and count the m marked ones.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator).

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0 otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2 (n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2 (n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

For the joint pdf $f(\mathbf{x} \mid \theta)$, $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ is the **likelihood function**; $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$ makes θ_1 more plausible. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2}\log 2\pi - \frac{1}{2}\sum_{i=1}^{n}(x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum(x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n\to\infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N(0, 1/I(\theta))$, i.e. $\hat{\theta}_n \approx N(\theta, \frac{1}{nI(\theta)})$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial\theta}\log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i = 1, \ldots, n_2$. Then

$$\hat{N}_{\text{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\text{MLE}}) \approx \frac{N^2(N - n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\text{MLE}}) = \frac{n_1^2 n_2(n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2}\Big(\frac{n_1}{N}\Big)^{X_i}\Big(1 - \frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m}\Big(1 - \frac{n_1}{N}\Big)^{n_2 - m},$$

$$\ell(N) = m\log n_1 - m\log N + (n_2 - m)\log(N - n_1) - (n_2 - m)\log N, \qquad \ell'(N) = \frac{n_2 - m}{N - n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N) = 0$: $N(n_2 - m) = n_2(N - n_1)$, so $\hat{N} = n_1 n_2/m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N}\log f(x \mid N) = \frac{1 - x}{N - n_1} - \frac{1}{N} = \frac{n_1/N - x}{N - n_1} = \frac{p - x}{N - n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p - X}{N - n_1}\Big)^2 = \frac{p(1 - p)}{(N - n_1)^2} = \frac{n_1}{N^2(N - n_1)}.$$

Hence $\mathrm{Var}(\hat{N}_{\text{MLE}}) \approx [n_2 I(N)]^{-1} = \frac{N^2(N - n_1)}{n_1 n_2}$. Plugging in $\hat{N} = n_1 n_2/m$:

$$\frac{(n_1 n_2/m)^2\,(n_1 n_2/m - n_1)}{n_1 n_2} = \frac{n_1^2 n_2^2}{m^2}\cdot\frac{n_1(n_2 - m)}{m}\cdot\frac{1}{n_1 n_2} = \frac{n_1^2 n_2(n_2 - m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\mathrm{Var}}(\hat{\tau}_y) \approx \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat{N} = 200 \cdot 100/20 = 1000$ and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{200^2 \cdot 100 \cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar{y}_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar{y}_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1 - \mu_2)$. ■

The bias is small only if N_2/N is small or $\mu_1 \approx \mu_2$; larger samples do not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1 - 1} \cdot \frac{N_1 - n_1}{N_1}}.$$

Only $0 \le p_2 \le 1$ is known. With $w_h = N_h/N$, $p = w_1p_1 + w_2p_2$; take $p_2 = 0$ and $p_2 = 1$:

$$\hat{p}_L = w_1\left(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\right) + w_2 \cdot 0,$$

$$\hat{p}_U = w_1\left(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\right) + w_2 \cdot 1.$$

If the interval covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p for any p_2, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: reducing nonresponse matters more than increasing n.

3 Selected exercises

Motivation. Lincoln–Petersen as completion of a 2×2 table and from the exact hypergeometric likelihood.

Exercise 10.1 (Deer population)

A random sample of $n_1 = 300$ deer is captured, tagged and released. A second sample of $n_2 = 200$ taken two weeks later contains $m = 62$ tagged deer.
(a) Arrange the data as a 2×2 table (caught or not in each sample) and estimate the missing cell, assuming independent captures.
(b) Show that the resulting estimate of N is the Lincoln–Petersen estimator, and place a bound on the error.

Solution.

(a) The table has one unobservable cell x (deer caught in neither sample):

	in 2nd sample	not in 2nd	total
in 1st sample	62	238	300
not in 1st	138	x	
total	200		N

Independence of the two captures means that the odds ratio of the table is 1, i.e. $62\,x = 238 \times 138$, so $\hat{x} = 32{,}844/62 = 529.7$.

(b)

Step 1. In general, with cells $n_{11} = m$, $n_{12} = n_1 - m$, $n_{21} = n_2 - m$ and $\hat{x} = n_{12}n_{21}/n_{11}$,

$$\hat{N} = n_{11} + n_{12} + n_{21} + \frac{n_{12}n_{21}}{n_{11}} = \frac{(n_{11}+n_{12})(n_{11}+n_{21})}{n_{11}} = \frac{n_1 n_2}{m}.$$

Step 2. Here $\hat{N} = 62 + 238 + 138 + 529.7 = 967.7 = 300 \times 200/62$.

Step 3. By the MLE theorem, $\widehat{\mathrm{Var}}(\hat{N}) = \dfrac{n_1^2 n_2 (n_2 - m)}{m^3} = \dfrac{300^2 \cdot 200 \cdot 138}{62^3} = 10{,}423$, and $2\sqrt{10{,}423} = 204$. The herd is estimated at 968 ± 204 deer. ■

Exercise 10.2 (The hypergeometric likelihood)

If the second sample is an SRS of n_2 drawn without replacement from N animals of which n_1 are marked, then

$$L(N) = P(m \mid N) = \frac{\binom{n_1}{m}\binom{N-n_1}{n_2-m}}{\binom{N}{n_2}}, \qquad N \geq n_1 + n_2 - m.$$

Assuming $m \geq 1$, show that the MLE of N over the integers is $\lfloor n_1 n_2 / m \rfloor$.

Solution. [Supplement]

Step 1. (Likelihood ratio.) For $N - 1 \geq n_1 + n_2 - m$, using $\binom{a}{c}/\binom{a-1}{c} = \frac{a}{a-c}$,

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-n_1}{n_2-m}}{\binom{N-1-n_1}{n_2-m}} \cdot \frac{\binom{N-1}{n_2}}{\binom{N}{n_2}} = \frac{N-n_1}{N-n_1-n_2+m} \cdot \frac{N-n_2}{N}.$$

Step 2. (Sign.) Both denominators are positive, so $L(N) \geq L(N-1)$ iff $(N-n_1)(N-n_2) \geq N(N-n_1-n_2+m)$. Expanding, the terms N^2 and $-N(n_1+n_2)$ cancel and the condition becomes $n_1 n_2 \geq Nm$, i.e. $N \leq n_1 n_2/m$.

Step 3. (Maximum.) L is nondecreasing for $N \leq n_1 n_2/m$ and strictly decreasing after, so maximal at $\lfloor n_1 n_2/m \rfloor$ (tied with $n_1 n_2/m - 1$ if integer), which is $\geq n_1 + n_2 - m$ since $(n_1 - m)(n_2 - m) \geq 0$: the Lincoln–Petersen estimate. ■

Chapter 11. Resampling Methods

1 Training Error and Test Error

Resampling redraws from the data to learn about an estimator: **cross-validation** and the **bootstrap** estimate prediction error, standard deviation and bias.

Definition 11.1

Let $D_n = \{(x_i, y_i)\}_{i=1}^n$ be the training data, g_{D_n} a prediction rule fitted to it, and L a loss function.

- The **training error** applies the fitted rule back to the training observations, e.g. $\frac{1}{n}\sum(y_i - \hat{y}_i)^2$.
- The **test error** (prediction error) is the expected loss on a new observation (x_0, y_0) independent of D_n:

$$PE = \mathrm{E}[L(y_0, g_{D_n}(x_0))].$$

The training error usually underestimates the test error (the rule is tuned to the same data): a curve through every training point has zero training error but predicts new $y_i^* = f(x_i) + \varepsilon_i^*$ badly. $MS_E = \frac{SS_E}{n-2}$ measures fit, not prediction.

With an independent test set $\{(x_j, y_j)\}_{j=1}^{n_t}$ we could estimate

$$\widehat{PE} = \frac{1}{n_t}\sum_{j=1}^{n_t} L(y_j, \hat{y}_j), \qquad \hat{y}_j = g_{D_n}(x_j),$$

but a large test set is often unavailable. The methods below hold out part of the training data instead.

2 Cross-Validation

2.1 K-fold cross-validation

Split the data randomly into K near-equal parts $C_1, \ldots, C_K$ (n_k observations, n/K if divisible); fit on the other $K-1$ parts and predict part k:

$$\mathrm{MSE}_k = \frac{1}{n_k}\sum_{i \in C_k}(y_i - \hat{y}_i)^2, \qquad CV_{(K)} = \sum_{k=1}^{K}\frac{n_k}{n}\mathrm{MSE}_k,$$

where $\hat{y}_i$ is the fit for observation i obtained with part k removed.

2.2 Leave-one-out cross-validation

$K = n$ gives LOOCV. Let $D_{(i)}$ be the training set with (x_i, y_i) omitted and $\hat{y}_{(i)} = g_{D_{(i)}}(x_i)$. Then

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_{(i)}),$$

which mimics $\widehat{PE}$ by letting each point serve as a test point for a rule that did not see it.

Advantages of LOOCV over a single validation split:

(i) much less bias, since each fit uses $n-1$ observations, almost the whole data set;

(ii) no randomness, since repeated runs give the same result.

Theorem 11.2: LOOCV shortcut for least squares

For least-squares linear regression $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$,

$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{y_i - \hat{y}_i}{1 - h_{ii}}\right)^2,$$

with full-fit $\hat{y}_i$ and leverage h_{ii}; $(y_i - \hat{y}_i)/(1-h_{ii}) = y_i - \hat{y}_{(i)}$ are the **PRESS residuals** (one fit).

Derivation: proof

Let $\mathbf{x}_i^\top$ be the ith row of $\mathbf{X}$, $A = \mathbf{X}^\top\mathbf{X}$, and $h_{ii} = \mathbf{x}_i^\top A^{-1}\mathbf{x}_i$. Deleting row i gives $A_{(i)} = A - \mathbf{x}_i\mathbf{x}_i^\top$ and, by the Sherman–Morrison formula,

$$A_{(i)}^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i^\top A^{-1}}{1 - h_{ii}}.$$

With $\mathbf{X}_{(i)}^\top\mathbf{y}_{(i)} = \mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i$,

$$\hat{y}_{(i)} = \mathbf{x}_i^\top A_{(i)}^{-1}(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i) = \left(\mathbf{x}_i^\top A^{-1} + \frac{h_{ii}\mathbf{x}_i^\top A^{-1}}{1-h_{ii}}\right)(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i) = \frac{\mathbf{x}_i^\top A^{-1}(\mathbf{X}^\top\mathbf{y} - \mathbf{x}_i y_i)}{1-h_{ii}}$$
$$= \frac{\hat{y}_i - h_{ii}y_i}{1-h_{ii}}.$$

Hence $y_i - \hat{y}_{(i)} = \dfrac{(1-h_{ii})y_i - \hat{y}_i + h_{ii}y_i}{1-h_{ii}} = \dfrac{y_i - \hat{y}_i}{1-h_{ii}}$.

Remark 11.3: Choice of K

LOOCV fits nearly identical training sets, so the errors are highly correlated and their average can be variable; $K = 5$ or 10 is usually better.

2.3 Why cross-validation works: model selection

Example 11.4: Choosing the degree of a polynomial

Consider $y_i = f(x_i) + \varepsilon_i$ fitted by $f_p(x) = \beta_0 + \beta_1 x + \cdots + \beta_p x^p$, and let $\hat{f}_p$ be its LS estimate. The goal is to select p. A natural target is

$$\mathrm{MSE}(p) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}(f(x_i) - \hat{f}_p(x_i))^2,$$

but f is unknown. Consider instead the prediction squared error for new responses $y_i^* = f(x_i) + \varepsilon_i^*$:

$$PSE(p) = \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}(y_i^* - \hat{f}_p(x_i))^2 = \frac{1}{n}\sum_{i=1}^{n}\mathrm{E}[(f(x_i) - \hat{f}_p(x_i)) + \varepsilon_i^*]^2$$
$$= \mathrm{MSE}(p) + \sigma^2,$$

because ε_i^* is independent of $\hat{f}_p$ and has mean zero. So minimizing PSE is equivalent to minimizing MSE. Now let

$$CV(p) = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{f}_p^{-i}(x_i)\right)^2,$$

where $\hat{f}_p^{-i}$ is fitted without the ith point. Since y_i is independent of $\hat{f}_p^{-i}$,

$$\mathrm{E}(y_i - \hat{f}_p^{-i}(x_i))^2 = \sigma^2 + \mathrm{E}(f(x_i) - \hat{f}_p^{-i}(x_i))^2,$$

and if $\hat{f}_p^{-i} \approx \hat{f}_p$, $\mathrm{E}[CV(p)] \approx \sigma^2 + \mathrm{MSE}(p) = PSE(p)$, so $\hat{p} = \arg\min_p CV(p)$ is sensible; the training error keeps decreasing in p.

3 The Bootstrap

The bootstrap quantifies the uncertainty of an estimator or a learning method, e.g. the standard error of a coefficient or a confidence interval for it.

3.1 A simple example

Invest α in an asset with return X and $1-\alpha$ in one with return Y, minimizing $\mathrm{Var}(\alpha X + (1-\alpha)Y)$.

Theorem 11.5

The risk is minimized by

$$\alpha = \frac{\sigma_Y^2 - \sigma_{XY}}{\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}}.$$

Proof. $g(\alpha) = \alpha^2\sigma_X^2 + (1-\alpha)^2\sigma_Y^2 + 2\alpha(1-\alpha)\sigma_{XY}$, and

$$g'(\alpha) = 2\alpha\sigma_X^2 - 2(1-\alpha)\sigma_Y^2 + 2(1-2\alpha)\sigma_{XY} = 0 \iff \alpha(\sigma_X^2 + \sigma_Y^2 - 2\sigma_{XY}) = \sigma_Y^2 - \sigma_{XY}.$$

Since $g'' = 2\mathrm{Var}(X-Y) \geq 0$, this is a minimum. ■

Plug in sample variances for $\hat{\alpha}$. In a simulation we could draw 1,000 data sets of 100 pairs, compute $\hat{\alpha}_1, \ldots, \hat{\alpha}_{1000}$, and report

$$\bar{\alpha} = \frac{1}{1000}\sum_{r=1}^{1000}\hat{\alpha}_r, \qquad \sqrt{\frac{1}{1000-1}\sum_{r=1}^{1000}(\hat{\alpha}_r - \bar{\alpha})^2}.$$

3.2 Back to the real world

With real data, the bootstrap instead resamples *the original data with replacement* to mimic new samples.

Definition 11.6: Bootstrap standard error

1. Draw Z^{*1} of size n from Z with replacement (some observations repeat, some are absent).
2. Compute the estimate $\hat{\alpha}^{*1}$ from Z^{*1}.
3. Repeat B times (say $B = 100$ or 1000) to get $\hat{\alpha}^{*1}, \ldots, \hat{\alpha}^{*B}$.

4. Estimate the standard error by

$$SE_B(\hat{\alpha}) = \sqrt{\frac{1}{B-1}\sum_{r=1}^{B}\left(\hat{\alpha}^{*r} - \bar{\alpha}^*\right)^2}, \qquad \bar{\alpha}^* = \frac{1}{B}\sum_{r'=1}^{B}\hat{\alpha}^{*r'}.$$

Example 11.7: Three observations

With $Z = \{(4.3, 2.4), (2.1, 1.1), (5.3, 2.8)\}$ (observations 1, 2, 3), possible bootstrap samples are $Z^{*1} = \{3, 1, 3\}$, $Z^{*2} = \{2, 3, 1\}$, $\ldots$, $Z^{*B} = \{2, 2, 1\}$ (by observation number), each yielding its own $\hat{\alpha}^{*r}$.

In R, `sample(x, replace=T)` draws a bootstrap sample of the indices.

4 Selected exercises

Motivation. The "63%" of a bootstrap sample (*ISLR*), the limit of the bootstrap SE, LOOCV as prediction-error estimate, and the optimism of the training error.

Exercise 11.1 (Which observations enter a bootstrap sample)

A bootstrap sample of size n is drawn with replacement from n observations. (After ISLR, Ch. 5, Exercise 2.)
(a) What is the probability that the first bootstrap observation is not the jth original observation? The second?
(b) Show that the probability that the jth observation is not in the bootstrap sample is $(1 - 1/n)^n$, and find its limit as $n \to \infty$.
(c) Find the probability that the jth observation is in the bootstrap sample for $n = 5, 100, 10{,}000$, and the expected number of distinct original observations in a bootstrap sample.

Solution.

(a) Each draw misses observation j with probability $1 - 1/n$; the second draw is independent and from the full set, so again $1 - 1/n$.

(b)

Step 1. The n draws are independent, so $P(j \notin Z^*) = \prod_{r=1}^{n} P(\text{draw } r \neq j) = (1 - 1/n)^n$.

Step 2. $\log(1-1/n)^n = n\log(1-1/n) = -1 - \frac{1}{2n} - \frac{1}{3n^2} - \cdots \to -1$, using $\log(1-u) = -u - u^2/2 - \cdots$ for $|u| < 1$. Hence $(1 - 1/n)^n \to e^{-1} = 0.368$.

(c)

Step 1. $P(j \in Z^*) = 1 - (1 - 1/n)^n$ equals $1 - 0.8^5 = 0.672$ for $n = 5$, $1 - 0.99^{100} = 0.634$ for $n = 100$ and 0.632 for $n = 10{,}000$, close to the limit $1 - e^{-1} = 0.632$.

Step 2. The number of distinct observations is $\sum_{j=1}^{n} \mathbf{1}\{j \in Z^*\}$, so by linearity its mean is $n\left[1 - (1 - 1/n)^n\right] \approx 0.632\, n$.

About 37% of the observations are left out of each bootstrap sample; these "out-of-bag" observations can serve as a test set for the rule fitted to that sample. ■

Exercise 11.2 (Ideal bootstrap standard error of the mean)

Let $x_1, \dots, x_n$ be fixed data and $\bar{x}^*$ the mean of a bootstrap sample $x_1^*, \dots, x_n^*$. Writing $\mathrm{E}^*, \mathrm{Var}^*$ for moments under bootstrap resampling, show that

$$\mathrm{E}^*(\bar{x}^*) = \bar{x}, \qquad \mathrm{Var}^*(\bar{x}^*) = \frac{\hat{\sigma}^2}{n}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2 = \frac{n-1}{n}s^2,$$

and conclude that $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ as $B \to \infty$.

Solution. [Supplement]

Step 1. The x_r^* are iid, each equal to x_i with probability $1/n$. Hence $\mathrm{E}^* x_r^* = \frac{1}{n}\sum_i x_i = \bar{x}$ and $\mathrm{Var}^* x_r^* = \frac{1}{n}\sum_i (x_i - \bar{x})^2 = \hat{\sigma}^2$.

Step 2. By independence, $\mathrm{E}^*\bar{x}^* = \bar{x}$ and $\mathrm{Var}^*\bar{x}^* = \hat{\sigma}^2/n$.

Step 3. The replicates $\bar{x}^{*b}$ are iid from the resampling distribution, so by the LLN $SE_B(\bar{x}) \to \hat{\sigma}/\sqrt{n}$ a.s.

This plug-in SE $\sqrt{(n-1)/n}\, s/\sqrt{n}$ is slightly below $s/\sqrt{n}$ (divisor n); finite-B Monte Carlo error is separate and controllable. ■

Exercise 11.3 (LOOCV for the sample mean)

Predict every response by the sample mean, $\hat{f} \equiv \bar{y}$.
(a) Show that $y_i - \bar{y}_{(i)} = \frac{n}{n-1}(y_i - \bar{y})$, where $\bar{y}_{(i)}$ is the mean without observation i, and hence $CV_{(n)} = \left(\frac{n}{n-1}\right)^2 \frac{1}{n}\sum_i (y_i - \bar{y})^2$. Check this against the LOOCV shortcut.
(b) If $y_1, \dots, y_n$ are iid with mean μ and variance σ^2, show that $\mathrm{E}\, CV_{(n)} = \sigma^2(1 + \frac{1}{n-1})$, the exact prediction error of a mean computed from $n-1$ observations.

Solution. [Supplement]

(a) $\bar{y}_{(i)} = \frac{n\bar{y} - y_i}{n-1}$ gives $y_i - \bar{y}_{(i)} = \frac{n(y_i - \bar{y})}{n-1}$; square and average. With $\mathbf{X} = \mathbf{1}$, $h_{ii} = \frac{1}{n}$, the shortcut gives the same $(y_i - \bar{y})/(1 - \frac{1}{n})$.

(b)

Step 1. $\mathrm{E}\sum_i (y_i - \bar{y})^2 = (n-1)\sigma^2$ for iid data (the unbiasedness of s^2). Hence $\mathrm{E}\, CV_{(n)} = \frac{n^2}{(n-1)^2} \cdot \frac{(n-1)\sigma^2}{n} = \frac{n\sigma^2}{n-1} = \sigma^2(1 + \frac{1}{n-1})$.

Step 2. For a new y_0 independent of a sample of size $n-1$ with mean $\bar{y}_{n-1}$, $\mathrm{E}(y_0 - \bar{y}_{n-1})^2 = \mathrm{Var} y_0 + \mathrm{Var}\bar{y}_{n-1} = \sigma^2 + \frac{\sigma^2}{n-1}$, the same value.

So LOOCV is exactly unbiased for the $n-1$-point rule and slightly pessimistic for the n-point rule (error $\sigma^2(1 + \frac{1}{n})$). ■

Exercise 11.4 (Optimism of the training error)

Let $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\mathbf{X}$ $n \times p$ of rank p, $\mathrm{E}\boldsymbol{\varepsilon} = \mathbf{0}$, $\mathrm{Var}\boldsymbol{\varepsilon} = \sigma^2\mathbf{I}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$, and $\mathbf{y}^* = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*$ new responses, $\boldsymbol{\varepsilon}^*$ an independent copy of $\boldsymbol{\varepsilon}$. Show that

$$\mathrm{E}\Big[\frac{1}{n}\|\mathbf{y} - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 - \frac{p}{n}\Big), \qquad \mathrm{E}\Big[\frac{1}{n}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2\Big] = \sigma^2\Big(1 + \frac{p}{n}\Big).$$

Solution. [Supplement]

Step 1. (Facts on $\mathbf{H}$.) $\mathbf{H}$ is symmetric and idempotent, $\mathbf{H}\mathbf{X} = \mathbf{X}$, and $\mathrm{tr}\mathbf{H} = \mathrm{tr}\big[(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\big] = \mathrm{tr}\mathbf{I}_p = p$. For a random vector $\mathbf{u}$ with mean $\mathbf{0}$ and $\mathrm{Var}\mathbf{u} = \sigma^2\mathbf{I}$ and a fixed matrix A, $\mathrm{E}(\mathbf{u}^\top A\mathbf{u}) =$

$\operatorname{tr}(A\,\mathrm{E}\mathbf{u}\mathbf{u}^\top) = \sigma^2 \operatorname{tr} A$.

Step 2. (Training error.) $(\mathbf{I}-\mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, so $\mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}-\mathbf{H})\boldsymbol{\varepsilon}$ and $\|\mathbf{y}-\hat{\mathbf{y}}\|^2 = \boldsymbol{\varepsilon}^\top(\mathbf{I}-\mathbf{H})\boldsymbol{\varepsilon}$. By Step 1 its mean is $\sigma^2(n-p)$.

Step 3. (Prediction error.) $\hat{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{H}\boldsymbol{\varepsilon}$, so $\mathbf{y}^* - \hat{\mathbf{y}} = \boldsymbol{\varepsilon}^* - \mathbf{H}\boldsymbol{\varepsilon}$ and

$$\mathrm{E}\|\mathbf{y}^* - \hat{\mathbf{y}}\|^2 = \mathrm{E}\|\boldsymbol{\varepsilon}^*\|^2 - 2\,\mathrm{E}\big(\boldsymbol{\varepsilon}^{*\top}\mathbf{H}\boldsymbol{\varepsilon}\big) + \mathrm{E}\big(\boldsymbol{\varepsilon}^\top\mathbf{H}\boldsymbol{\varepsilon}\big) = n\sigma^2 - 0 + p\sigma^2,$$

the middle term vanishing by independence and $\mathrm{E}\boldsymbol{\varepsilon}^* = \mathbf{0}$.

Step 4. Divide by n. The gap $2p\sigma^2/n$ grows with p: training error falls with model size while prediction error eventually rises, so select models by CV (or a penalty $2p\hat{\sigma}^2/n$). ■